

A Shortfall in Investor Expectations of Leveraged Tokens

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Abstract

Leveraged tokens (LVTs) are emerging crypto-assets primarily issued by centralized exchanges. The concept is borrowed from leveraged ETFs (LETFs) in traditional financial markets, which offer higher gains (and higher losses) relative to price movements in the underlying asset. Leverage is commonly used by short-term traders to amplify returns from daily market shifts. However, LVTs have been implemented differently from LETFs by exchanges in the crypto market, with variations across platforms. We examine the mechanics and constituent components of LVTs, demonstrating that the lack of a standard has resulted in deficiencies and unexpected technical and economic outcomes. To identify existing problems, we analyze more than 1,600 leveraged tokens from 10 issuers. Our analysis reveals that 99.9% of LVTs are centralized, with 80% lacking blockchain interaction, leading to transparency issues. Total supply information is difficult to access for 53% of them, and 41% appear inadequately backed at launch. Additionally, 97% of LVTs are vulnerable to front-running during well-known events, and they deviate from their stated leverage ratios more than LETFs, partly due to inconsistent re-leveraging processes and higher management fees. This work provides a framework for crypto investors, blockchain developers, and data analysts to gain a deep understanding of leveraged tokens and their impact on market dynamics, liquidity, and price movements. It also offers insights for crypto exchanges and auditors into the internal functionalities and financial performance of LVTs under varying market conditions.

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1 Introduction

A typical Exchange-Traded Fund (ETF) is a weighted basket of stocks from firms with a common characteristic (e.g., they all operate in a specific sector or have a high market capitalization). The issuer splits the basket into shares, which are bought and sold on exchanges just like individual stocks [48].

► **Example 1.** One of the most widely traded ETFs is the *SPDR S&P500 ETF*, with the



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ticker symbol *SPY*. It is issued by SSGA¹ and holds a basket of stocks from nearly 500 publicly traded companies that are included in the S&P500² index. The S&P500 index has globally served as a gauge for the performance of the U.S. stock market as a whole, due to its depth and diversity. Since *SPY* tracks the S&P500 index, investors can gain broad exposure and diversify their investment risk across the stock performance of 500 companies in 11 sectors without the logistics or starting capital required to buy shares in all these companies.

Leveraged ETFs (LETFs) were introduced in 2006 and are ETFs designed to amplify the daily performance of the underlying basket (more on leverage in Section 3.1).³ Inverse LETFs aim to achieve a return that is a multiple of the inverse of the underlying asset's daily performance [39, 20, 63]. Many investors alternatively refer to LETFs and inverse LETFs as “Bullish” and “Bearish” LETFs, respectively, reflecting their short-term sentiment on future price movements.

► **Example 2.** *Direxion Daily S&P500 Bull 3x ETF (SPXL)* is a 3x (three times) LETF that seeks to deliver triple the daily performance of the S&P500. It magnifies each 1% gain in the S&P500 index into a 3% gain and loses 3% for every 1% drop in the index. *Direxion Daily S&P500 Bear 3x ETF (SPXS)* delivers triple the opposite daily performance of the S&P500 index. If the S&P500 index depreciates by 1%, *SPXS* gains 3%, and vice versa [77, 46].

A Leveraged Token (LVT) in the cryptocurrency and crypto-asset (“crypto”) market can be compared to a Leveraged ETF (LETF) in the traditional financial market. Similar to LETFs, LVTs use leveraged products available in the crypto market to outperform the underlying asset's return on a daily basis. While the majority of LETFs are actively managed funds⁴, LVTs employ one of three management models: (i) centralized, (ii) decentralized, and (iii) hybrid. Centralized LVTs are primarily managed by crypto exchanges and can be purchased on the spot market or directly from the exchange (*cf.* Appendix A.5 on investing in LVTs). Decentralized LVTs operate on-chain and can be traded by interacting directly with the smart contract. Hybrid LVTs are essentially decentralized LVTs that are traded on centralized crypto exchanges. Users trade on centralized exchanges for their user-friendly interfaces, continuous-time order books (rather than automated market makers, which are the only trading mechanism efficient enough to run on-chain), and increased liquidity due to aggregated buy and sell orders.⁵ However, this model introduces certain disadvantages resulting from the combination of centralized and decentralized systems (*e.g.* functional complexities, security concerns, custodial risks, *etc.*).

► **Example 3.** An issuer may offer *BTC3L/BTC3S* as a pair of LVTs tracking Bitcoin (BTC) as the underlying asset. A Bitcoin futures contract (*BTC-Perp*⁶) can be used as the leveraged product to outperform Bitcoin in the short term. The number three in the LVT name

¹ State Street Bank and Trust Company (SSGA) is one of the three dominant companies in the ETF market, with a 14.01% market share, following BlackRock and Vanguard, which have 33.64% and 29.16%, respectively [68].

² The S&P 500 index comprises 500 of the top publicly traded companies in the U.S. It was launched in 1957 by the credit rating agency Standard and Poor's [41].

³ The underlying asset can be stocks, market indexes (*e.g.*, S&P 500, NASDAQ-100, *etc.*), commodities (*e.g.*, gold, oil, corn, *etc.*), or any asset with a price.

⁴ In actively managed funds, investment managers actively buy and sell assets with the goal of outperforming a specified benchmark index, resulting in higher management fees.

⁵ The more liquid an asset is, the easier and more efficient it is to convert back into cash. Less liquid assets take more time and may incur higher costs [37].

⁶ A type of Bitcoin futures contract without a defined expiration date (known as a “perpetual” contract).

■ **Table 1** *Left table*: Number of issued leveraged tokens per year, average per year, and number of unique underlying assets, which we collected manually from different sources. An underlying asset might be used to create multiple tokens with different leverage levels. *Right table*: Characteristics of issued LVTs by different issuers. Only 20% of tokens have been created on the blockchain. 99.9% of LVTs use derivatives as the leveraged product, which is offered by the same issuer (Internal for Pionex as of Jan 2023). Except for *Index Coop*, the rest of the issuers use off-chain fund management systems. Rebalancing triggers for *Index Coop* are still off-chain.

Issuer	Number of LVTs per year							Underlying Assets
	2019	2020	2021	2022	2023	Total	Average	
MEXC		162	116	102	76	456	114	217
AscendEX				228	112	340	170	94
Gate.io		116	96	54	8	274	69	123
Pionex		60	102	36	2	200	50	76
FTX	102	27				129	65	43
KuCoin			50	6	38	94	31	45
Binance		38	2			40	20	20
ByDFi				16	24	40	20	20
ByBit				34		34	34	17
Index Coop			2			2	2	2
Total	102	403	368	476	260	1609	322	654

Characteristics of LVTs			
Fund Source	Blockchain Rep.	Fund Management Algorithm	Leveraged Product
Internal	No	Off-Chain	Futures
Internal / External	Yes		
Internal	No		
External	Yes	On-Chain / Off-Chain	Debt

represents the multiplier (triple-leveraged), while L/S stands for going long/short on the market.⁷ BTC3L gains 3% when the price of Bitcoin rises by 1%, and loses 3% for every 1% price drop. Conversely, when Bitcoin drops by 1%, BTC3S gains 3%, and loses 3% for every 1% price rise.

Since 2019, more than 1,600 LVTs have been issued by various crypto exchanges. The FTX exchange introduced the original concept by issuing 102 tokens on the blockchain.⁸ Trading volumes exceeded \$1 million per day [24]. This upward trend has continued, with other exchanges issuing approximately 32 new LVTs per month on average from January 2020 to November 2023 (see Table 1).

Motivation for studying LVTs.

- *LVT attractiveness for investors*: Investment in LETFs nearly doubled in 2022 compared to 2021 [70], demonstrating an appetite for low-risk leverage, which is satisfied in the crypto market by LVTs. LVTs reduce liquidation risks compared to derivatives and margin trading. However, other characteristics (e.g., volatility drag) must be understood to avoid unexpected value destruction. These risks are not unique to LVTs; they also exist in LETFs.⁹ (See Appendix A.5, and comparative Table 6 for more details on why investors are attracted to this type of token.)
- *LVT distinctive dynamics*: Section 2 offers a cohesive framework for understanding key aspects of LVTs, such as their underlying dynamics, peculiarities in product design, effects on crypto markets, and investor suitability. Leveraged products can impact market dynamics, especially in highly volatile markets [66]. More technical details are provided in Section 3, which can be useful for those involved in the design and implementation of LVTs to understand how these tokens affect liquidity and price movements, potentially influencing the robustness and reliability of trading algorithms.

⁷ Going long refers to buying an asset with the expectation that its value will increase, allowing it to be sold for a profit later. Conversely, going short refers to profiting from a decline in the asset's value [43].

⁸ The ERC-20 standard is the most prominent standard for fungible tokens on the Ethereum blockchain. These tokens can represent financial assets such as LVTs and can be exchanged between users.

⁹ In 2018, Credit Suisse had to close an ETF ETN after its price plunged 90% in one day. In another example, WisdomTree had to close its 3x oil products in March 2020 after their value was wiped out [70].

- *Regulatory implications:* LVTs introduce new risks for market regulation, investor protection, and financial stability. Our work contributes to broader discussions on how to effectively regulate emerging financial technologies like LVTs. Additionally, since LVTs are often held by commercial firms requiring audited financial statements [26], auditors should understand how LVTs function, their risks, and how they perform under different market conditions.

Contributions of this paper.

In this work, we study more than 1,600 leveraged tokens from 10 issuers, examining various aspects such as underlying assets, interaction with the blockchain, types of leveraged products, and fund management algorithms. We dedicate part of the paper to carefully explaining LVT mechanics and constituent components to help the reader understand the functionality of leveraged funds, rebalancing mechanisms, and smart contracts. We then address six research questions about LVTs:

- RQ 1: What information is visible to traders of an LVT?
- RQ 2: To what extent are LVTs locked to the offering exchange?
- RQ 3: Are the LVTs offered today adequately backed?
- RQ 4: What are the possibilities of front-running in LVTs?
- RQ 5: How well do LVTs track their asserted leverage ratios?
- RQ 6: Are LVT fees in-line with traditional LETFs?

Methodology of measurements and dataset.

To extract the list of issuers, we identified the top 100 crypto exchanges based on 24-hour trading volume, as reported by crypto comparison websites.¹⁰ Then, by visiting each provider’s website, we manually checked whether LVTs were offered. We supplemented this with searches on Google¹¹, online forums and blogs¹², and crypto news sites¹³. The combined list should be comprehensive as of 2023 and includes LVTs from various types of exchanges (both large and small, centralized and decentralized), covering different asset classes.

The majority of LVTs exhibited common elements that allowed for formal representation in Section 2. We reviewed each LVT’s documentation to understand its functionality [33, 8, 25, 57, 61, 45, 11, 16, 15, 23], enabling us to identify their components as discussed in Section 3. Outliers that did not fit into the typical LVT model due to unique structural features were not excluded (e.g., Hybrid LVTs by FTX). Instead, the model was generalized to include these 6% of outliers. We have described the parts of the model that were extended for this category (e.g., the smart contract component for decentralized LVTs).

In Section 4, the functionality of tokens is evaluated by answering six research questions. To address these, we collected data from both the token issuers and historical data available through various exchanges and financial databases. Issuer documentation provided crucial details on the structure, mechanics, and intended use of LVTs. Historical data were gathered from reputable financial data providers¹⁴, including price histories, trading volumes, issuance dates, and other relevant metrics. Additionally, we cross-referenced data from multiple sources

¹⁰ Websites: coinecko.com, coinmarketcap.com, cryptocompare.com, and coinranking.com.

¹¹ Search terms: leveraged tokens, leveraged ETFs cryptocurrency, leveraged crypto assets, crypto leverage trading platforms, crypto leverage trading token issuers, etc.

¹² Crypto blogs and forums such as reddit.com (e.g., /cryptocurrency or /binance subreddits).

¹³ News sources: coindesk.com and cointelegraph.com.

¹⁴ Data sources: tradingview.com, cryptodatadownload.com, etherscan.io, and finance.yahoo.com.

to validate the accuracy and consistency of our dataset. It is worth mentioning that the analyzed data from sources (e.g., tradingview.com) are aggregated directly from the source exchanges, ensuring the information is accurate and up-to-date. The direct connection to exchanges means that the data reflect real-time market conditions and historical performance accurately. Moreover, these data sources are widely used and trusted within the financial and cryptocurrency communities, providing tools and data to a large number of traders, analysts, and researchers, underscoring their credibility.

Related work.

To our knowledge, this is the first academic paper on LVTs; however, our work overlaps with studies on LETFs, which have established the following research findings:

- The effect of compounded returns intensifies with longer holding periods, causing LETFs to struggle to maintain their stated leverage over time. As a result, long-term performance is not linearly related to the return of the underlying asset [72, 51, 36, 21, 18].
- LETFs can underperform over longer periods without efficient rebalancing. Researchers state that frequent rebalancing during periods of high volatility is necessary to maintain leveraged exposure to the tracking index. They conclude that reducing rebalancing frequency can significantly decrease tracking errors [18, 17, 27].
- The impact of LETFs on market volatility and liquidity shows that their daily rebalancing can increase volatility and trading volume near the market close, potentially distorting the market price of LETFs and creating additional inefficiencies [21, 36, 18, 71, 62, 66].
- Investors often do not fully understand the mechanisms, risks, and proper uses of LETFs, which require tolerating increased risk. Consequently, LETFs are suitable only for experienced and skilled investors who comprehend the complexities and hazards of trading with them [18, 27, 47].

2 Price and Return Dynamics

As daily returns is embedded in the design of LVTs, a k -leveraged LVT should generally earn k times of the daily return of the underlying. The amplification ratio, known as leverage (k), can be fixed or dynamic. A proportional change in the underlying price is k -times the proportional change in LVT price. In the short-term, LVT return is consistent with k , but beyond a single day, the return is highly path dependent, making LVTs unsuitable for buy-and-hold strategies. This is an issue that is ignored by most retail investors, leading to unexpected loss of capital. In the following, the price and return dynamics of LVTs have been discussed, aiding analysis and simulation of such issues.

2.1 LVT Price Dynamics

LVTs are intentionally designed with leverage as a core component of their architecture. They are aimed at outperforming the return of the underlying benchmark on a daily basis.¹⁵ Let P_{t_n} represent the LVT price at calendar time t_n , expressed as:

$$P_{t_n} = P_{t_{n-1}} \left(1 + k \frac{\Delta S_{t_n}}{S_{t_{n-1}}} \right) \quad n \in [1, 365), t \geq 0, k \in [-5, -0.5] \cup [0.5, 5] \quad (1)$$

¹⁵ Returns will be slightly lower after deducting fund management fees, accounting for market volatility, interest paid on borrowing, and other associated expenses.

S_{t_n} is the underlying price at time t_n , indexed by n , where n denotes the days of the year. The frequency of n does not have to be daily; it can be redefined in hours or minutes without any loss of generality. However, since daily returns are embedded in the LVT product design, n is effectively daily. P_{t_n} represents the price of the LVT at the close of trading day n . $S_{t_{n-1}}$ and $P_{t_{n-1}}$ are the initial prices of the underlying asset and LVT, respectively, at the beginning of trading day n (or at the end of trading day $n-1$). ΔS_{t_n} is the amount of change in the underlying price relative to the initial price. The constant variable k is the LVT multiplier (leverage), which can be defined as either a fixed or dynamic value, depending on the issuer. LVTs with fixed leverage can take values from the set $\{-5, -3, -2, -1, -0.5, 0.5, 2, 3, 5\}$, while dynamic leverage fluctuates within the range $[-4.0, -1.25] \cup [1.25, 4.0]$.

The multiplier k further divides LVTs into three main functional groups: (i) Long LVTs, where $k \in \{2, 3, 5\} \cup [1.25, 4.0]$. The value of a long LVT rises k times faster than the underlying asset and is profitable in rising markets; (ii) Short LVTs, where $k \in \{-5, -3, -2, -1\} \cup [-4.0, -1.25]$. The value of a short LVT rises $|k|$ times faster than the underlying asset and is profitable in falling markets. This type of LVT is also used to hedge¹⁶ positions or as a substitute for short-selling the underlying asset; (iii) Low-risk LVTs, where $k \in \{-0.5, 0.5\}$. Low-risk LVTs can be used to reduce the impact of adverse market movements without the full risks associated with higher leverage factors. Every 1% change in the underlying leads to a 0.5% change in the price of a low-risk LVT (*i.e.* less profit with less risk). Equation (1) indicates a linear relationship between the LVT price and the underlying price from time t_{n-1} to t_n . It can be shown algebraically that:

$$P_{t_n} = P_{t_{n-1}} + k P_{t_{n-1}} \left(\frac{\Delta S_{t_n}}{S_{t_{n-1}}} \right) \Rightarrow \frac{\Delta P_{t_n}}{P_{t_{n-1}}} = k \left(\frac{\Delta S_{t_n}}{S_{t_{n-1}}} \right) \quad (2)$$

When n and $n-1$ are close enough to each other in equation (2), the proportional change in the LVT price relative to its initial price equates to the proportional change in the underlying price relative to its initial value. Equation (2) then becomes:

$$\frac{dP_t}{P_t} = k \frac{dS_t}{S_t} \quad (3)$$

If there is not enough initial capital for the fund, the issuer may borrow $(k-1)P_t$ from external financial sources at an interest rate of $r \geq 0$. Considering f as the expense ratio of the fund, equation (3) can be completed as:

$$\frac{dP_t}{P_t} = k \frac{dS_t}{S_t} - ((k-1)r + f)dt \quad (4)$$

As will become clear later in Section 2.2, equation (4) does not represent the return of an LVT; rather, it only shows the LVT price change relative to the underlying for discrete time intervals. Discrete time is commonly used in financial modeling, where prices are calculated at specific intervals, such as daily, hourly, or minutely. Therefore, equation (1) can be used, for example, to calculate the LVT price change on trading day 2 compared to day 1. The price of an LVT at any arbitrary interval can also be modeled in continuous time, as detailed in Appendix A.12. However, for the modeling of LVTs in this work, discrete time is more appropriate, reducing unnecessary complications.

¹⁶ Hedging is a common practice to reduce the risk of adverse price movements in another position. For example, opening a \$100K long Bitcoin position and hedging it by buying a \$20K 2x short Bitcoin LVT. If the long position experiences a 5% price drop, the net loss is partially offset by gains from the hedge position, resulting in a net loss of $(2 \times 5\% \times \$20K) - (5\% \times \$100K) = \$2K - \$5K = -\$3K$, which is less than the $-\$5K$ loss without hedging.

2.2 LVT Return Dynamics

The return of LVTs cannot simply be considered as k times the return of the underlying asset. Let $R_{t_{(n-1)} \rightarrow n}$ represent the return of the underlying asset (in percentage) at price S from time t_{n-1} to t_n :

$$R_{t_{(n-1)} \rightarrow n} = \frac{\Delta S_{t_n}}{S_{t_{n-1}}} \quad n \geq 1, t \geq 0, t_0 = 0 \quad (5)$$

Considering the relationship between the price of LVT and the underlying in (3), the return of the LVT from time t_{n-1} to t_n can be expressed as $G_{t_{(n-1)} \rightarrow n} = 1 + kR_{t_{(n-1)} \rightarrow n}$. Since daily return is embedded in the design of LVTs, the frequency of n here is daily. In the short run (a trading day), where the underlying volatility is almost constant, the change in the LVT price relative to the underlying can be assumed to be equivalent to k -times the return. However, in the long run (several trading days or weeks), LVT return may be significantly lower or higher than the underlying due to the compounding effect, as given by:

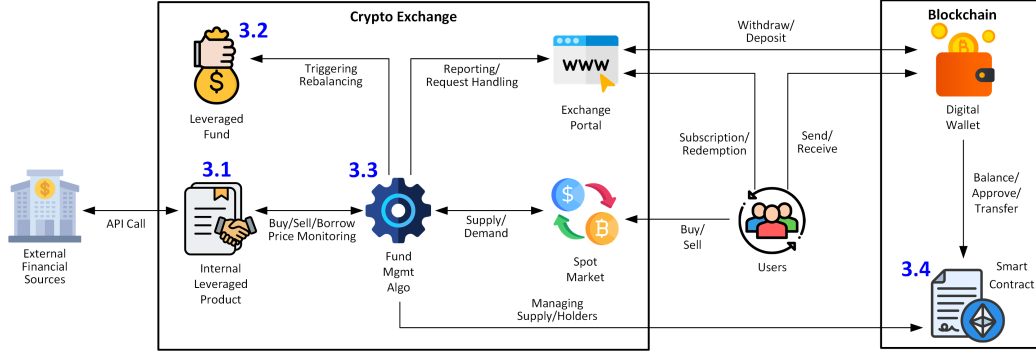
$$G_{t_0 \rightarrow n} = (1 + kR_{t_0 \rightarrow 1})(1 + kR_{t_1 \rightarrow 2}) \cdots (1 + kR_{t_{(n-1)} \rightarrow n}) = \prod_{n=1}^N (1 + kR_{t_{(n-1)} \rightarrow n}) \quad (6)$$

The longer-term return of LVTs is impacted by the carried-over k multiplier on each trading day, which magnifies both profit and loss due to the compounding effect. Even though the intention behind long and short LVTs is to move in opposing directions on a daily basis, it is common for both types to generate negative cumulative returns when held over an extended period (see example 30 in Appendix A.5 on the compounding effect).

Takeaways: Daily returns are embedded in the design of LVTs. Generally, a k -leveraged LVT should earn k times the daily return of the underlying asset. The amplification ratio, known as leverage (k), can be fixed or dynamic. Leverage divides LVTs into three functional groups: Long, Short, and Low-risk. A proportional change in the underlying price results in a k -times proportional change in the LVT price. In the short term, LVT returns align with k , but beyond a single day, the return becomes highly path-dependent, making LVTs unsuitable for buy-and-hold strategies. This is an issue ignored by most retail investors, leading to unexpected capital losses.

3 Leveraged Token Mechanics

LVTs are tokenized representations of a leveraged fund whose value is derived from the value of a leveraged product. Leveraged products are essential components of LVTs, allowing issuers to form a leveraged fund and offer it as centralized or decentralized tokens. 99.9% of LVT issuers use crypto futures as the leveraged product. To properly reflect the value of the fund through circulating LVTs, the notional value of all tokens must match the fund's notional value. As the price of the leveraged product changes over the trading day, the leverage of the fund gradually diverges from the promised ratio. The *fund management algorithm* resets this deviation by buying or selling the leveraged product on a daily basis. It also implements the logic of the LVT and defines how it should function in different market conditions. If an LVT is designed to be fully on-chain and decentralized, the smart contract provides the functionality of the *fund management algorithm* as well, by extending common features of the ERC-20 standard. The constituent components of LVTs vary depending on the issuer. According to issuer documentation, the general components of a typical LVT are illustrated in Figure 1, followed by a brief explanation of the functionality of each component.



■ **Figure 1** The constituent components of LVTs, according to the issuer’s documentation. Some issuers have implemented them internally, resulting in missing blockchain components.

3.1 Leveraged product

Financial managers of LETFs use leverage to open positions worth more than the required capital. In LVTs, leveraged exposure cannot appear out of nowhere. In the absence of Total Return Swaps (TRS) in the crypto market (more on TRS and financial leverage in Appendix A.1), other mechanisms for achieving leverage are (i) opening positions in the crypto derivatives market, which provides up to 200x leverage, and (ii) borrowing capital from external sources, generating up to 10x leverage. LVT issuers typically do not use high multiples, offering tokens with up to 5x leverage. This allows them to choose either derivatives or debt as the leveraged product. 99.9% of issued LVTs use futures (a type of derivative), and only *Index Coop* uses the debt market to finance investments.¹⁷ The desired outcome for *Index Coop*’s tokens is to generate future returns that outweigh the cost of borrowing. Other issuers that use derivatives aim to minimize dependency on other exchanges for buying and selling futures. They often offer the corresponding futures trading in their own portfolio to facilitate LVT management and reduce the cost of transactions between exchanges (compare the *Leveraged Product* and *Fund Source* columns in Table 1). For example, every issuer that launched BTC Long/Short tokens offers BTC-Perp futures as the underlying. Internal leveraged products facilitate LVT operations, such as adjusting fund positions, monitoring underlying price fluctuations, and triggering fund rebalancing.

3.2 Leveraged fund

It is a fund that derives its *notional value* from a basket of leveraged products.¹⁸ The leveraged products provide leveraged exposure, upon which the value of the issued tokens is based. Let V_{t_n} represent the price of the k -times leveraged product V on trading day n , tracking the underlying asset S . The price of V_{t_n} differs from S as it carries k -times exposure. A leveraged fund L with a notional value of L_{t_n} can be formed by purchasing a basket of V , given by:

$$L_{t_n} = kV_{t_n}B_{t_n}(1 + (\rho_{t_n} + \phi_{t_n})) \quad t \geq 0, 0\% \leq \rho_{t_n} + \phi_{t_n} \leq 0.5\%, \rho_{t_0} = 0, \phi_{t_0} = 0 \quad (7)$$

¹⁷ *Index Coop* uses money market protocols on Ethereum (e.g., Compound protocol) that offer permissionless borrowing and lending capabilities [22].

¹⁸ The notional value represents the total value of a financial instrument or contract at its full face value (i.e., controlled money by the financial instrument). The notional value is not typically exchanged between counterparties; instead, it serves as a reference point for calculating payments or obligations [31].

Where B_{t_n} is the number of V units forming the fund at the rebalancing time t_n . ρ_{t_n} and ϕ_{t_n} represent management and futures funding fees, respectively. ρ_{t_n} is always negative, as the issuer deducts associated expenses from the fund's value. ϕ_{t_n} can be positive or negative, depending on the *futures funding fee* payments (more in Section 4.4.3). The sum of ρ_{t_n} and ϕ_{t_n} (i.e., the total daily fee) varies per LVT issuer and ranges from 0.01 to 0.5 percent daily. Note that the change in the price of the leveraged product (V_{t_n}) is proportional to the underlying price (S_{t_n}), but does not vary based on the k multiplier. More precisely, $V_{t_n} = V_{t_{n-1}}(1 + R_{t_{(n-1) \rightarrow n}})$. In equation (7), L_{t_n} represents the financial value controlled by the leveraged fund L , which originates from the leveraged product V . The change in the price of V is proportional to S , but the value of L changes with respect to the k factor. In simple terms, V represents the price of the leveraged product, while L represents the amount of money that can be controlled using V .

► **Example 4.** An issuer may arrange an *Ether long double-leveraged fund* by purchasing 4 *Ether-Perp long 2x futures* at \$1.5K (V_{t_0}). The 2x leverage of Ether-Perp allows the issuer to pay half of the Ether price, which is assumed to be \$3K (S_{t_0}). With zero fees in (7) at t_0 ($\rho_{t_0} + \phi_{t_0} = 0$), a leveraged fund L worth $2 \times \$1.5K \times 4 = \$12K$ can initially be formed (L_{t_0}). A 10% change in the price of Ether affects the price of futures by the same 10%, bringing it to \$1.65K (V_{t_n}). However, the notional value of the fund (L_{t_n}) changes according to $k = 2$, reaching $2 \times \$1.65K \times 4 = \$13.2K$. This demonstrates the effect of k on L compared to V .

LVTs are issued with a certain initial supply that can be adjusted through the *Subscription* and *Redemption* process (more on this process in Appendix A.5). For added or removed tokens, the issuer offsets the notional value of the fund with the notional value of the tokens by buying or selling the corresponding amount of the leveraged product. Let N_{t_n} represent the total supply of a k -leveraged LVT at time t_n . The notional value of the issued LVTs (A_{t_n}) can be expressed as:

$$A_{t_n} = kN_{t_n}P_{t_n} \quad (8)$$

Equating (7) and (8) gives the total number of tokens (N_{t_n}) that should exist at the price of P_{t_n} at time t_n . Mathematically, this can be expressed as: $kN_{t_n}P_{t_n} = kV_{t_n}B_{t_n} \Rightarrow N_{t_n} = \frac{V_{t_n}B_{t_n}}{P_{t_n}}$, $t \geq 0, P_{t_0} \geq 1$.

► **Example 5.** Assume $P_{t_0} = \$10$ as the initial offering price of ETH2L in the previous example 4. P_{t_0} is typically set by the issuer at either \$1 or \$10 per token.¹⁹ The initial supply of ETH2L at \$10 per token is:

$$N_{t_0} = \frac{V_{t_n}B_{t_n}}{P_{t_n}} = \frac{\$1.5K \times 4}{\$10} = 600$$

The notional value of all 600 tokens ($A_{t_0} = 2 \times 600 \times \$10 = \$12K$) is consistent with the notional value of the leveraged fund ($L_{t_0} = 2 \times \$1.5K \times 4 = \$12K$). Investors purchase a portion of this fund in the form of LVT, allowing them to generate twice the profit compared to the underlying Ether-Perp. Essentially, the value of the ETH2L token is derived from the *Ether long 2x leveraged fund*, which in turn is derived from the 4 positions in the *Ether-Perp long 2x futures*.

¹⁹ Taken from the initial public offering (IPO) price of a SPAC (Special Purpose Acquisition Company), which is typically set at a nominal \$10 per unit. Unlike a traditional IPO, the SPAC IPO price is not based on the valuation of an existing business but rather on future income expectations.

3.3 Fund management algorithm

As the price of the leveraged product (V_{t_n}) fluctuates over time, the notional value of the leveraged fund (L_{t_n}) changes, causing the leverage ratio of the LVT to deviate from the stated leverage. Let $\tilde{k}_{t_n}$ represent the realized leverage ratio that the notional value of the tokens (A_{t_n}) represents at time t_n , expressed as $\tilde{k}_{t_n} = \frac{L_{t_n}}{N_{t_n} P_{t_n}}$.

► **Example 6.** Referring to the first trading day of ETH2L in examples 4 and 5, in which the price of Ether (S_{t_0}) increases by 10%, the notional value of the fund (L_{t_n}) changes to $2 \times \$1.65K \times 4 = \$13.2K$. The 2x leverage of the LVT increases its price by 20%, rising to \$12 (P_{t_n}) from the initial \$10 (P_{t_0}). Since the LVT supply remains constant at 600 tokens, the leverage ratio of the fund drops from 2x to 1.8x. ($\tilde{k}_{t_n} = \frac{L_{t_n}}{N_{t_n} P_{t_n}} = \frac{\$13.2K}{600 \times \$12} = 1.8\bar{3}$).

The analysis above suggests that with the change in the price of the underlying (S_{t_n}) and subsequently the price of the leveraged product (V_{t_n}), the notional value of the leveraged fund (L_{t_n}) changes, and the realized leverage ratio of LVTs ($\tilde{k}_{t_n}$) becomes higher or lower than the stated leverage k . Mathematically, if $\mathbb{E}[\tilde{k}_{t_n}]$ represents the expected change in $\tilde{k}_{t_n}$ in relation to the underlying price change (S_{t_n}), applying equations (1) gives us:

$$\mathbb{E}[\tilde{k}_{t_n}] = \frac{k V_{t_n} B_{t_n} (1 + R_{t_{(n-1)} \rightarrow t_n})}{k N_{t_n} P_{t_n}} = \frac{V_{t_n} B_{t_n} (1 + R_{t_{(n-1)} \rightarrow t_n})}{N_{t_n} P_{t_{n-1}} (1 + k R_{t_{(n-1)} \rightarrow t_n})} \quad (9)$$

As the number of tokens (N_{t_n}) remains constant while the price of the underlying changes at a rate of $(1 + R_{t_{(n-1)} \rightarrow t_n})$, the denominator of (9) changes k -times faster (or slower) than the numerator, resulting in positive or negative leverage skewness. This highlights the need to re-leverage the fund on a daily basis, a process managed by the *fund management*.²⁰

Fund management is an off-chain algorithm (or on-chain for decentralized tokens) that dynamically adjusts the fund to maintain the leverage at the expected ratio. When the token's leverage increases, it sells some of the fund's positions to reduce the leverage and return it to the expected level (*cf.* Appendix A.7 for rebalancing details). The majority of algorithms are off-chain with no interaction with the blockchain. The only on-chain instance is implemented by *Index Coop* [22]. In addition to correcting the leverage, the algorithm interacts with other components to adjust supply, update balances, monitor the price of the underlying, and deduct daily fees.

3.4 On-chain contracts

For decentralized LVTs, a smart contract represents the leveraged fund on the blockchain. It is typically implemented as an ERC-20 token [30, 7, 64], allowing users to exchange LVTs on the blockchain without issuer intervention. As indicated in the *Blockchain Representation* column of Table 1, 80% of issuers have not created LVTs on the blockchain. As a result, this component is missing from Figure 1. The absence of a smart contract leads to several deficiencies, which are discussed in the next section.

Takeaways: An LVT is a tokenized representation of a leveraged fund, whose value is derived from the value of a leveraged product. 99.9% of LVT issuers use crypto futures as the leveraged product. As the price of the leveraged product fluctuates over a trading day, the leverage of the fund gradually diverges from the promised ratio. The *fund management algorithm* resets this deviation by buying or selling the leveraged product on a daily basis.

²⁰ Also referred to as *fund management agent*, *fund management party*, or *certified fund manager*.

4 Research Questions

Due to the lack of a common standard in LVTs for defining the rebalancing process, data transparency, implementation standards, etc., these tokens are issued with varying features at the discretion of the issuer. We examine the characteristics of issued tokens per issuer and discuss the respective deficiencies as research questions RQ1 to RQ6.

4.1 RQ1: What information is visible to traders of an LVT?

Among the 10 LVT issuers, only *FTX* and *Index Coop* have created tokens on the Ethereum blockchain. *FTX*'s management model was hybrid (*i.e.* tradable decentralized tokens on a centralized exchange), while only *Index Coop*'s tokens are fully decentralized. The remaining 8 exchanges prefer to implement LVTs centrally and entirely internal.

► **Example 7.** Binance leveraged tokens (BLVTs) are one of the centralized LVTs that are entirely accessible within Binance's ecosystem. They can be exclusively traded on Binance's spot market with no possibility of withdrawal. BLVTs are not even published on Binance's own blockchain (BNB Smart Chain) and are created more like a pseudo-crypto.²¹

4.1.1 Transparency in total supply

Total supply is used to calculate the Net Asset Value (NAV) of LVTs as a representation of the market's fair value. Due to imbalances in supply and demand, the market price of LVTs may deviate from the NAV, trading at a premium or discount. In the long run, LVT prices converge to the NAV due to a mechanism similar to arbitrage in traditional markets. Orders placed far from the NAV price lose or gain value over time (*cf.* Appendix A.13 on the general arbitrage mechanism). In the short run, however, investors use the NAV as a reference price when buying or selling, especially in bulk. The NAV of LVTs can be calculated by equating (7) and (8), with the current token supply N_{t_n} :

$$kN_{t_n}P_{t_n} = kV_{t_n}B_{t_n} \Rightarrow P_{t_n} = \frac{V_{t_n}B_{t_n}}{N_{t_n}} \quad t \geq 0, N_{t_0} \geq 1 \quad (10)$$

► **Example 8.** In the previous examples (4) to (6), when the Ether price increases by 10% on day 2, the market price of ETH2L trades at \$12 (after a $2 \times 10\% = 20\%$ increase), while its NAV price is $(\$1.65K \times 4)/600 = \11 . ETH2L is, in fact, overvalued, and traders should wait for either (i) the arbitrage mechanism to play out and bring the LVT price down, or (ii) the next rebalancing schedule, which will match the fund's value with the notional value of the tokens.

For LVTs hosted on the blockchain, total supply is public and can be retrieved for NAV calculations. However, for centralized LVTs, investors must refer to the exchange's website. The total supply of tokens on some exchanges, such as AscendEX, Pionex, Gate.io, and ByDFi, does not appear to be public, making it difficult to verify the real value of LVTs (*i.e.* 53% of all tokens). LVTs are *open-end funds* with a theoretically unlimited token supply.²² Issuers can increase the supply based on market liquidity and demand for the

²¹ A cryptocurrency that is not sufficiently decentralized [4].

²² Open-end funds can issue an unlimited number of shares. The fund sponsor sells shares directly to investors and redeems them as well. The NAV per share of an open-end fund is calculated daily by dividing the total value of the fund (minus liabilities) by the total number of shares outstanding [19].

token. Transparency in the number of issued tokens builds trust and reduces the risk of investment. Moreover, it addresses audit questions such as, Has the fund's value changed proportionately after increasing or decreasing the supply of LVTs? How much were the fund's value deviations in the previous audit period, and were they within the acceptable range?

4.1.2 Transparency in transactions

Transactions on the blockchain show the flow of tokens and the movement of the fund. This enables investors to analyze transactions and ensure the expected functionality of LVTs.

► **Example 9.** We reviewed all *Mint* and *Burn* transactions of ETCBULL (FTX 3x Long Ethereum Classic) on the blockchain.²³ The analysis suggests that a total of 51,640,895 tokens were issued, and 24,207 were destroyed (*i.e.* 51,616,688 circulating tokens). The trend of issuing tokens has taken on exponential velocity since April 2022. A total of 783,022 tokens were issued during the 960-day period between October 2019 and May 2022, while 50,857,873 tokens were issued over just 184 days from April to October 2022. In other words, 98.5% of all tokens were issued in just 6 months. Checking the recipient address indicates FTX's possible sub-wallet as the receiver. This sudden change in token supply warrants further investigation, especially given FTX's collapse shortly afterward. A possible explanation for this anomaly is presented in Appendix A.11.

This is just an example indicating the importance of transparency in LVT transactions. Transactions of centralized tokens are not public and only available to the issuer. Statistically, transactions of 80% of LVTs cannot be analyzed as we did in the above case.

4.1.3 Transparency in token holders

Holders of tokens created on the blockchain are public, allowing investors to check them as a measure of the token's liquidity. A small number of market participants reduces the token's liquidity and can make it more challenging to execute large orders. It may also lead to a wider bid-ask spread, increasing the cost of executing trades. Investors generally prefer assets with higher liquidity, narrower bid-ask spreads, and more market participants.

► **Example 10.** 90% of XRPBULL (FTX 3x Long Ripple) tokens are distributed among four holders.²⁴ In another example, three accounts own 94% of all issued FTX 3x Long Cardano (ADABULL) tokens.²⁵

Holding a large number of tokens by a limited number of accounts can noticeably elevate investment risk. One holder may decide to sell a significant number of tokens at any moment, potentially resulting in a notable price drop within a short period of time, leading to significant losses for smaller holders. Since most issuers do not publish the list and respective ownership percentages of centralized LVTs, the participants of 80% of LVTs remain uncertain.

4.1.4 Inability to audit

Conducting audits ensures the security, functionality, and compliance of LVTs as claimed by the issuer. Unlike centralized LVTs, the code of tokens created on the blockchain is public,

²³ ETCBULL transactions on the Ethereum blockchain filtered for issued and deposited tokens to FTX's own address: <https://bit.ly/3MwHVqv>.

²⁴ List of XRPBULL holders: <https://bit.ly/3MQurrc>.

²⁵ List of ADABULL holders: <https://bit.ly/3MUxPRZ>.

allowing auditors to identify vulnerabilities and associated risks. The security of these 20% decentralized LVTs can be evaluated by reviewing the code against industry best practices such as SWC [67]. Moreover, external audits are essential for LVTs to ensure they function as intended, such as verifying the output of methods when transferring tokens or updating balances. Auditors may also provide recommendations to improve the security, functionality, and compliance of LVTs. For centralized LVTs, the code is not public, requiring cooperation and willingness of the issuer to conduct a thorough review and quality assessment. Sharing the code and the results of an independent audit would improve transparency and help build trust between token holders and issuers.

Takeaways: In some centralized LVTs, investors do not have access to crucial information such as total supply, transactions, and holders. Total supply is a necessary parameter for calculating the fair value of tokens and is also used by auditors to evaluate the consistency and efficiency of LVTs. Transparency in transactions enables investors, auditors, and anyone involved in the LVT ecosystem to analyze token flow, detect suspicious activities, enhance security, ensure compliance, and verify token functionality. The number of holders for centrally issued tokens is unknown, leading to a much higher investment risk compared to decentralized counterparts.

4.2 RQ2: To what extent are LVTs locked to the offering exchange?

4.2.1 Interoperability with dapps and DeFi

In 2019, the total value locked in Decentralized Finance (DeFi) was approximately 700 million USD. As of April 2022, it stands at around 150 billion USD, representing more than 200% growth in less than three years [76]. Hosted LVTs on the blockchain (which usually comply with one of the fungible²⁶ token standards) facilitate interaction with DeFi systems, unlocking potential interoperability opportunities.

► **Example 11.** FTX was able to employ blockchain interoperability to share its ETHBULL (3x Long Ethereum) with other exchanges such as Poloniex, Indodax, Bittrex, and Gate.io.²⁷ These exchanges owned 20%, 4%, 3%, and 2% of ETHBULL, respectively, and offered it on their platforms due to the possibility of interaction with DeFi.

In contrast, centrally issued LVTs cannot interact with other platforms and operate in isolation, preventing LVTs from moving across different platforms and systems. Decentralized LVTs, on the other hand, foster connectivity and enable users to access a wide range of services and functionalities without being confined to a single exchange.

4.2.2 Inability to custody

At first glance, the custody issue seems common to all assets on centralized exchanges. However, BTC buyers can transfer it to their personal wallets, while centralized LVTs remain locked within the exchange. Holders do not own the actual tokens but are simply betting on price movements. Some explain this custodial issue by viewing LVTs as “token contracts”, though this term is not widely recognized nor aligns with the functionality of crypto derivatives. LVTs are essentially tokenized forms of derivative exposures.

²⁶ Fungible (interchangeable) token standards are widely used by decentralized applications (dApps) to interact with other applications. ERC-20 is the dominant standard, followed by ERC-777 and ERC-1155.

²⁷ List of ETHBULL holders: <https://bit.ly/3MSZX7P>.

► **Example 12.** BTCUP and BTCDOWN are issued by Binance and track Bitcoin as the underlying asset. Unlike Bitcoin holders, owners of these tokens cannot withdraw or transfer them to their own digital wallets. In contrast, similar Bitcoin leveraged tokens were created by FTX on the Ethereum blockchain (known as BULL and BEAR tokens). Holders of these tokens still had the opportunity to exchange them on decentralized exchanges, such as Uniswap²⁸, shortly after FTX's bankruptcy. Holders could recover 80% of the token value on the first day of the bankruptcy, 50% on the second day, and up to 20% on the third day.

Takeaways: The inability to self-custody centralized LVTs raises greater concerns compared to decentralized counterparts, potentially increasing the investment risk in these crypto-assets. Another important advantage of tokens created on the blockchain is their interoperability with other dApps, crypto exchanges, and the DeFi ecosystem in general. LVTs that interact with DeFi offer the opportunity to participate in a more open and transparent financial system that operates without the need for intermediaries.

4.3 RQ3: Are the LVTs offered today adequately backed?

The simplest definition of an LVT is a tokenized leveraged fund. According to the documentation of LVT issuers, 99.9% of leveraged funds derive their value from a basket of positions in the futures market [8, 33, 11, 15, 16, 61, 25, 45, 57]. The issuer must either (i) offer futures trading in their portfolio, or (ii) open futures positions on other crypto exchanges and manage them systematically through APIs as the underlying asset fluctuates.²⁹ The question we raise, due to the lack of external audits, is to what extent LVT issuers have properly prepared futures contracts before launching LVTs. Have users invested in tokens that are properly backed, or are they simply trusting the issuer and potentially investing in tokens with no real value?

4.3.1 Missing futures product

Some issuers have launched LVTs without offering the corresponding futures products. While we cannot rule out the possibility that they hold the necessary futures positions on other exchanges, this raises concerns that these LVTs might not be adequately backed.

► **Example 13.** AscendEX uses its own futures products and does not rely on futures products from other exchanges. However, they issued 3x/5x Long/Short Monero (XMR3L/S and XMR5L/S) without offering XMR perpetual contracts initially. To our knowledge, no XMR futures products were available on the market from any exchange to be used as leveraged products at the time of the launch of these XMR tokens.

The above example is one of 390 issued tokens lacking a corresponding futures product (see column B of Table 2). Based on available historical data and information from the issuer's website, 24% of LVTs did not have the necessary futures product offered by the same issuer and instead relied on futures from other exchanges.

²⁸ DEX transactions of FTX 3X Long Bitcoin Token (BULL) and FTX 3X Short Bitcoin Token (BEAR) on the Ethereum blockchain: <https://bit.ly/3p0h3uz>, <https://bit.ly/41Da181>.

²⁹ Among 10 LVT issuers, Pionex used the *Binance Broker API* for its *Futures Arbitrage Bot*, but it has been terminated since June 2021 [56]. After reviewing Pionex's documentation [57], it remains unclear whether *Binance Futures* is still used as the leveraged product for LVTs. However, Pionex launched its own futures product in January 2023. If they no longer use *Binance Futures* and rely solely on their own futures product, it appears that 148 LVTs did not have corresponding futures contracts at the time of launch (e.g., ETC3L/S, ZRX2L/S, XLM3L/S).

■ **Table 2** *Left table*: Number of issued LVTs with delayed or missing futures products, analyzed using historical data and undisclosed information. To our knowledge, 41% of the issued LVTs did not have sufficient financial backing at the time of launch. *Right table*: Rebalancing and fee deduction schedules, which we collected manually from the issuers' websites. Regardless of leverage type, rebalancing is performed daily at different times. Additionally, Threshold-Based (TBR) or Out of Range (OOR) rebalancing methods are used to trigger interim rebalancing. Fund expenses are also deducted at various times with variable percentages.

Issuer	Delayed Futures Launch	Missing Futures Product	Total Delayed or Missing	Total Launched LVTs	% of unbacked LVTs
	(A)	(B)	(C)	(D)	(E)
AscendEX	36	214	250	340	74%
Pionex	0	148	148	200	74%
MEXC	176	12	188	456	71%
ByDFi	16	0	16	40	40%
FTX	18	0	18	129	14%
Gate.io	18	16	34	274	12%
Binance	0	0	0	40	0%
ByBit	0	0	0	34	0%
KuCoin	0	0	0	94	0%
Index Coop	0	0	0	2	0%
Total	264	390	654	1609	41%

Leverage	Rebalancing Schedule		Fee deduction	
	Regular Daily	Interim	Daily Schedule	Expense Ration
(a)	(b)	(c)	(d)	(e)
Fixed	02:30 UTC	10% TBR / OOR	00:00 UTC	0.500%
Fixed / Variable	00:00 UTC+8	10% TBR	00:00 UTC+8	0.030%
Fixed	00:00 UTC	15% TBR	00:00 UTC+8	0.100%
	08:00 UTC+8			0.030%
	02:00 UTC	10% TBR	00:00 UTC	0.030%
	00:00 UTC-4		00:00 UTC+8	0.300%
Variable	N/A	10% TBR / OOR	00:00 UTC	0.010%
Fixed / Variable	00:00 UTC			0.005%
Fixed / Variable	08:00 UTC+8	14% TBR	23:45 UTC+8	0.045%
Fixed	00:00 UTC	20% TBR	00:00 UTC	0.023%

4.3.2 Delayed futures product

Missing futures products are not the only issue with centralized LVTs. For 264 tokens, the corresponding futures product was only offered after the issuance of the tokens (see column A of Table 2). In other words, at the time of the launch of 17% of LVTs, the required futures may not have existed. According to the LVT documentation on issuers' websites, these issuers did not disclose using futures from other crypto exchanges. Internal futures trading was introduced later, after the token was launched.

► **Example 14.** MEXC issued 3x Long/Short Cardano (ADA3L/S) in February 2020, while ADA-Perp was only launched in July 2020, resulting in a 154-day delay. They did not disclose using futures from other exchanges, indicating the fund might have been operating without financial backing during this period.

According to available information, on average, 41% of LVTs have missing or delayed futures products (see column E of Table 2). The main financial issue with LVTs is the lack of transparency in the fund management system. Centralized LVTs function like a black box to investors and are fully managed by the issuer. Even for tokens with proper futures backing (e.g., Binance, ByBit, and KuCoin), investors can rely solely on numeric assertions made on the issuer's website.

Takeaways: The value of LVTs is derived from a leveraged fund, which itself is based on the value of futures. In the absence of futures at the time of offering, investors may have purchased LVTs that lacked proper financial backing. An analysis of available historical data on the issuer's website shows that, on average, 2 out of every 5 issued LVTs did not appear to have proper financial backing at the time of launch. Although the exchange may have addressed this issue over time, the required futures contracts were missing at the time of issuance. Without external audits, investors can rely solely on the exchange's claims and assume that LVTs are financially backed as promised.

4.4 RQ4: What are the possibilities of front-running in LVTs?

Front-running is an illegal practice in the equity market where non-public information is used to purchase shares of a company before the price moves [29, 78, 9]. For instance,

FINRA³⁰ announced a \$700K fine against *Citadel Securities*³¹ in 2020 for front-running activities between 2012 and 2014 [54]. In the design of LVTs, certain well-known events can be exploited by traders to benefit from anticipated price movements. They can engage in similar front-running practices that may impact the price of the underlying asset and the token itself. We review these events and explore possible front-running scenarios as follows.

4.4.1 Event I: Impending fund rebalancing

Rebalancing is the process of maintaining the desired leverage ratio of funds over time. To keep the leverage at the stated ratio, issuers perform periodic rebalancing. This can be triggered at predefined intervals (e.g., every day, every 8 hours, or every n blocks), or upon meeting certain conditions (e.g., after exceeding a specific threshold). All LVT issuers perform regular daily rebalancing and trigger interim rebalancing in volatile markets (see columns B and C of Table 2). They may trigger rebalancing when the underlying asset's price fluctuates by more than $X\%$, or when the leverage passes a threshold. The *fund management algorithm* governs the rebalancing process, adjusting futures positions and restoring the leverage ratio to the target level (*cf.* Appendix A.7 for details on the rebalancing process).

The number of contracts that must be bought or sold to restore the leverage is predictable, making front-running possible. Let ΔB_{t_n} represent the number of required futures contracts to rebalance the fund at time t_n . ΔB_{t_n} can be easily calculated by considering the return of the underlying asset from time t_{n-1} to t_n ($R_{t_{(n-1) \rightarrow n}}$). The number of required futures contracts to restore the fund leverage can be calculated by subtracting the notional value of the tokens (equation 8) from the notional value of the fund (equation 7):

$$\Delta B_{t_n} = (1 + kR_{t_{(n-1) \rightarrow n}})kN_{t_{n-1}}P_{t_{n-1}} - (1 + R_{t_{(n-1) \rightarrow n}})kV_{t_{n-1}}B_{t_{n-1}} - (\rho_{t_n} + \phi_{t_n})L_{t_n} + \epsilon_{t_n}$$

This equation is quadratic and can be simplified as $ax^2 - bx - c$ for long tokens, and $-ax^2 + bx - c$ for short tokens. When the underlying return is positive ($R_{t_{(n-1) \rightarrow n}} > 0$), ΔB_{t_n} is always positive ($a > 0$), and when the return of the underlying is negative, ΔB_{t_n} is negative as well ($a < 0$). In simpler terms, for long LVTs, futures exposure must be increased when the underlying price is rising, and decreased when the underlying price is falling (see *Fund Basket Delta* in Tables 12 and 13 of appendix A.7).

There is also a fund expenses term $(\rho_{t_n} + \phi_{t_n})L_{t_n}$, which is usually deducted from the fund's value to cover operating expenses. However, this term can turn positive when the received funding fees (ϕ_{t_n}) exceed the fund expenses (ρ_{t_n}). ϵ_{t_n} represents a disturbance term that captures the effects of news or shocks in the underlying. Since rebalancing is a predictable event, by buying or selling ΔB_{t_n} of the leveraged product, other traders can front-run the trade, potentially impacting the price of the token or even the underlying asset.

► **Example 15.** Consider the following sequence in which Alice calculates ΔB_{t_n} to potentially front-run the rebalancing trade:

1. Alice checks the issuer's website for the upcoming rebalancing of BTC5L (5x Long Bitcoin). She notices the next daily rebalancing is scheduled for 00:00 UTC.
2. Alice calculates the number of contracts that will be bought or sold by the issuer to maintain the 5x target leverage of BTC5L.

³⁰ The Financial Industry Regulatory Authority (FINRA) is a government-authorized organization that oversees U.S. equity markets by regulating member brokerage firms and exchange markets.

³¹ Citadel Securities is the largest designated market maker on the New York Stock Exchange (NYSE).

3. Alice front-runs the rebalancing trade by placing an order just before 00:00 UTC (ahead of the rebalancing trade). If she anticipates that the algorithm will buy Bitcoin futures, she may buy Bitcoin futures expecting increased demand, driving up the price, and giving her the opportunity to sell futures at higher prices. Conversely, if BTC5L will be selling the fund's positions, leading to increased supply, she may sell Bitcoin futures.

Front-running in the above example may not only manipulate the price of Bitcoin futures but also inflate the price of BTC5L. Consider the following scenarios (A) and (B), with corresponding calculations in Table 8 of appendix A.5.

- Alice calculates the *Basket Delta* of BTC5L prior to the rebalancing schedule and realizes that the algorithm will purchase 594.21 new contracts at \$33,000 (Scenario A in Table 8 of appendix A.5). Assuming this purchase increases the price of Bitcoin futures by 1%, she can buy contracts just before the rebalancing trade at \$33,000 and sell them afterward at \$33,330. A 1% increase in the underlying price inflates the token price to \$15.05. This provides Alice an additional opportunity to buy the token for \$15 before the rebalancing and sell it at a higher price afterward.
- The effect of Alice's strategy in the previous scenario may be amplified if many traders engage in front-running. The increased demand may raise the price of Bitcoin futures even before the rebalancing trade. If the influx of other traders pushes Bitcoin futures up by 1%, and the rebalancing trade further increases the price by another 1%, this secondary effect could also inflate the token price further and create more price distortion (Scenario B in Table 8 of appendix A.5).

4.4.2 Event II: Management fee deduction

Similar to LETFs, daily fees and expenses are deducted from the leveraged fund to cover associated costs (*cf.* Appendix A.8 on incurred costs). The fee rate and daily schedule vary by issuer (see columns D and E of Table 2). Management fees are deducted at specific times, allowing adversaries to exploit this known event, potentially coinciding with rebalancing. The simultaneous occurrence of events I and II can intensify the front-running effect during the rebalancing process.

► **Example 16.** Consider the same sequence as the previous example, where Alice calculates the *Basket Delta* of BTC5L at the same time as the management fee deduction. The issuer's withdrawal of \$99,000 (Management fee row in Table 8 of appendix A.5) reduces the fund's value. To compensate, $\$99,000/\$33,000 = 3$ additional contracts need to be purchased. The coincidence of these two events causes the rebalancing algorithm to slightly increase demand by purchasing 597.21 contracts instead of 594.21.

4.4.3 Event III: Futures funding fee exchanges

Funding fee is a mechanism in Perps to converge the price of contracts with the price of the underlying crypto. It is calculated based on the notional value of the futures position and is exchanged between short and long traders who keep their positions open. Shorts pay longs when the funding rate is negative, and longs pay shorts when the rate is positive (*cf.* Appendix A.9 for details on funding fee dynamics). The intervals for *Funding fee exchange* are public and displayed on the issuer's website, typically occurring every 8 hours at 00:00 UTC, 08:00 UTC, and 16:00 UTC. Since the fund is composed of futures, it either pays or receives funding fees at these times. This predictably increases or decreases the value of the leveraged fund, which can be exploited to amplify the effects of front-running.

The impact of front-running can be exacerbated when events I, II, and III occur simultaneously. Such concurrency may force the algorithm to buy or sell more contracts than would be required for fund rebalancing alone (*i.e.* $\Delta \tilde{B}_{t_n} = \sum_{n=1}^3 \Delta B_{t_n}$).

► **Example 17.** As calculated in the *Basket Delta* row of Table 8 in appendix A.5, in the first rebalancing cycle, an additional 2.70 futures contracts are required to cover the 0.3% daily management fee deduction and 0.03% daily funding fee exchange. This means that 2.7 more contracts will be added if events II and III coincide with event I. The impact on basket delta can be even more significant as the value of the fund increases in a volatile market (see *Fee Basket Size* row in Table 12 of appendix A.5 on rebalancing in volatile markets).

To mitigate front-running in LVTs, issuers should avoid rebalancing the fund on predetermined schedules. Techniques such as intraday or randomized rebalancing, or algorithmic trading, can help reduce the visibility of rebalancing trades.³² Only Binance avoids regular daily rebalancing, instead triggering it when the underlying price fluctuates more than 10% or when leverage falls outside the range of $[-4.0, -1.25] \cup [1.25, 4.0]$. This means 97% of current LVTs perform fund rebalancing at specific intervals, increasing the likelihood of daily front-running (see column B of Table 2).

Takeaways: The possibility of front-running in LVTs arises from the predictability of impending fund rebalancing, management fee deductions, and futures funding fee exchanges. Adversaries can exploit the temporary changes in supply and demand initiated by the fund. This effect may be intensified if all three events occur simultaneously or if there is large-scale participation by multiple traders. Front-running has also been a concern in LETFs, though it is mitigated by the continuous oversight of regulatory bodies such as the SEC and FINRA.

4.5 RQ5: How well do LVTs track their asserted leverage ratios?

The leverage ratio of LVTs is determined by the issuer and can be either variable (dynamic) or fixed. If an LVT uses the *Underlying+Leverage+Long/Short* naming convention, the leverage is most likely fixed. The *Underlying+Up/Down* format is used for LVTs with variable leverage.

► **Example 18.** KuCoin has issued ETH3L as a 3x long token tracking Ether as the underlying. Binance similarly offers ETHUP and ETHDOWN tokens with a target leverage in the range of $[1.25, 4]$ and $[-4, -1.25]$, respectively.

Very high leverage factors such as $\pm 10x$ or $\pm 15x$ are not common in currently issued LVTs, as the majority of them provide $\pm 3x$ leverage (see Figure 3a in Appendix A.4). Low leverage is aimed at minimizing losses and extending the liquidation point during periods of high volatility. Highly leveraged LVTs lose value in the same proportion as the underlying asset and may not be attractive to investors. Crypto exchanges advertise LVTs as an investment vehicle providing leveraged exposure to crypto-assets with minimal liquidation risk. However, LVTs with high leverage factors defeat this promise.

³² Iceberg orders, which are large orders broken into smaller lots, are a sophisticated trading algorithm used to execute rebalancing trades in smaller, more discrete chunks over time.

4.5.1 Inconsistency of fixed leverage

Approximately 16% of LVTs are issued with variable leverage, fluctuating in the range of $[-4.0, -1.25] \cup [1.25, 4.0]$. Additionally, 9%, 59%, and 11% of LVTs have fixed 2x, 3x, and 5x leverage ratios, respectively. As shown in Table 10 of Appendix A.6, LVTs with fixed leverage may not always provide exactly the promoted leverage. One aspect of risk involves leverage deviation (also called tracking error). Furthermore, some issuers do not rebalance the fund in the same way.

► **Example 19.** MEXC and KuCoin adjust the leverage of only the tokens that have lost value. Consider a volatile market where Bitcoin loses 10% in a day. These issuers adjust only the leverage of BTC3S, while the leverage of BTC3L remains unchanged, as it gained value [33, 44]. For example, if the price of Bitcoin is \$30K and the fund holds 600 contracts, the fund's initial value is \$18M. Assuming 600K issued tokens at an initial offering price of \$10, the target leverage is 3x (*i.e.* $k = (600 \times \$30K) / (600K \times \$10) = 3$). A 10% increase in Bitcoin's price changes the leverage of BTC3L and BTC3S to 2.53x and 4.71x, respectively. However, these exchanges correct the leverage of BTC3S to prevent further capital loss in case of more price decline. As a result, this rebalancing process undervalues the BTC3L fund (*i.e.* inflates the value of BTC3L). Instead of only rebalancing the losing side, both BTC3L and BTC3S positions should be adjusted simultaneously to bring the leverage back to 3x as advertised by the issuer.

We compared the leverage deviation of Bitcoin LVTs with LETFs over the course of a year. Analysis details are provided in Appendix A.6 and Table 10. As can be seen, LVTs exhibit higher leverage deviations than similar products in the equity market. This issue becomes more apparent when comparing the standard deviation of returns in the equity and crypto markets. Leverage deviation leads to underperformance or overperformance of tokens, causing investors to experience returns that deviate from the intended amplification effect of LVTs. This is particularly important in light of previous research on LETF returns, which shows that LETFs, on average, do not negatively impact investor short-term returns [49]. Results indicate that the daily return distribution using real-world historical data is significantly more leptokurtic than the normal distribution. However, in LVTs, returns tend to have a wider or flatter shape (platykurtic) due to higher leverage deviation.

4.5.2 Disadvantages of variable leverage

LVTs with variable leverage aim to: (i) minimize the impact of volatility drag (*cf.* Appendix A.10), and (ii) reduce the possibility of front-running (as discussed in Section 4.4). LVTs are advertised as an investment vehicle that amplifies returns relative to a certain multiplier, although this factor changes constantly in tokens with variable leverage. This introduces an additional risk dimension, requiring regular monitoring and adjustment of positions as the leverage fluctuates. Additionally, these types of tokens rebalance on an as-needed basis with no predetermined schedules. Rebalancing can therefore be triggered by (i) a sudden fluctuation in the underlying price (such as more than 15%), (ii) exceeding the expected leverage range (such as above 4x or below 1.25x for long LVTs), and (iii) handling subscription or redemption requests, which change the total supply. One disadvantage of this type of rebalancing is that funds can remain undervalued or overvalued for extended periods.

► **Example 20.** The rebalancing events of BTCUP (a Long Bitcoin LVT by Binance) over the past 3 years are listed in Table 14 of Appendix A.6. A rebalancing event occurred on 03-Jan-2023, 204 days after the previous one on 13-Jun-2022. During those 204 days, no

rebalancing event was triggered because the changes in Bitcoin's price did not exceed the 10% threshold limit, and the fund's leverage fluctuated within the expected range of 1.25x to 4x. During this period, the fund's value was much lower (or higher) than the amount required to support the value of issued tokens, but no rebalancing occurred. Undervalued funds may benefit the issuer, while investors hold inflated tokens.

Another disadvantage of dynamic leverage is the imbalance in rebalancing triggers. Some issuers initiate the rebalancing process at different ranges for long and short tokens.

► **Example 21.** Pionex triggers rebalancing when the leverage of long tokens exceeds the range of [2.2, 4.0], while this range is [1.8, 4.8] for short tokens [55]. This inconsistency increases the complexity of position management, leading to unfavorable outcomes for investors.

LVTs with variable leverage may reduce the probability of front-running but also reduce token transparency and increase the complexity of managing positions. In the absence of a specific standard for the rebalancing of dynamic leverage, each issuer implements its own algorithm, which is not necessarily consistent with others. This causes confusion for investors when switching from one exchange to another, as they may expect similar performance.

Takeaways: As the price of the underlying asset fluctuates, the value of the fund and the presented leverage change at different rates. The price of the token might be at a premium or discount, not reflecting the actual value of the fund. All LVT issuers reconcile the total value of the tokens and the fund through a daily rebalancing schedule. However, the way this process is implemented differs by issuer, leading to deviations from the target leverage. Rebalancing in both types of tokens—those with fixed and dynamic leverage—has its shortcomings. Referring to LETFs, where almost 100% of traditional LETFs adhere to a fixed leverage strategy [69], suggests that fixed leverage may be more appropriate for LVTs as well. However, there is still a need to reduce the current high leverage deviation compared to LETFs, which can be achieved by modifying the relevant algorithms.

4.6 RQ6: Are LVT fees in-line with traditional LETFs?

Issuers of LETFs/LVTs charge daily fees to cover the associated costs of operating the fund (*cf.* Appendix A.8 on associated expenses). In recent years, fees have generally come down, with the average annual Management Expense Ratio (MER) for traditional ETFs and LETFs being 0.45% and 0.95%, respectively. We reviewed the daily fees of all LVTs and summarized the results in Table 3. Depending on the issuer, the annual MER for LVTs varies from 1.83% to 36.5%. Comparatively, the standard deviation of MER in LETFs and LVTs is 0.38% and 34.21%, respectively. This signifies that fees are less predictable and more volatile in LVTs than in LETFs (up to 90 times), impacting the overall expense ratio and net returns. Additionally, high fees often indicate a developing market with a lack of a broad base of issuers. This could be a risk indicator for investors, who typically expect lower fees due to increased competition and improved market maturity.

Takeaways: High daily fees in LVTs act as a continuous drag on performance, eroding returns, leading to underperformance, and making them less attractive. Daily costs in LVTs should be lower than in LETFs, given that futures transactions are internal and there is much less regulatory overhead. Furthermore, higher fees serve as a risk indicator for developing markets with limited competition and inefficient cost management.

Table 3 Comparison of the daily expense ratio in the equity and crypto markets. Each day, the daily fee is deducted from the price of ETFs/LETfs/LVTs, which negatively impacts the ROI. Therefore, investing in assets with lower daily rates (green rows) is less risky with higher return.

Market	Symbol	Issuer	Underlying Index/Asset	Leverage	Annual Expense Ratio	Daily Expense Ratio	Market	Symbol	Issuer	Underlying Index/Asset	Leverage	Annual Expense Ratio	Daily Expense Ratio
Equity	IVV	BlackRock	S&P500	+1x	0.0300%	0.000119%	Crypto	BTC3L	ByBit	Bitcoin	+2x to +4x	1.8250%	0.005000%
	VOO	Vanguard	S&P500	+1x	0.0300%	0.000119%		BTC3S	ByBit	Bitcoin	-2x to -4x	1.8250%	0.005000%
	SPY	SSGA	S&P500	+1x	0.0945%	0.000375%		BTCUP	Binance	Bitcoin	+1.25 to +4x	3.6500%	0.010000%
	QQQ	Invesco	NASDAQ-100	+1x	0.2000%	0.000794%		BTCDOWN	Binance	Bitcoin	-1.25 to -4x	3.6500%	0.010000%
	TQQQ	ProShares	NASDAQ-100	+3x	0.8600%	0.003413%		BTC3L	Pionex	Bitcoin	+2.2x to +4x	10.9500%	0.030000%
	SH	ProShares	S&P500	-1x	0.8800%	0.003492%		BTC3S	Pionex	Bitcoin	-2.2x to -4x	10.9500%	0.030000%
	SSO	ProShares	S&P500	+2x	0.8900%	0.003532%		BTC3L	ByDFi	Bitcoin	+3x	10.9500%	0.030000%
	SDS	ProShares	S&P500	-2x	0.9000%	0.003571%		BTC3S	ByDFi	Bitcoin	-3x	10.9500%	0.030000%
	SPXU	ProShares	S&P500	-3x	0.9000%	0.003571%		BTC3L	KuCoin	Bitcoin	+3x	16.4250%	0.045000%
	UPRO	ProShares	S&P500	+3x	0.9100%	0.003611%		BTC3S	KuCoin	Bitcoin	-3x	16.4250%	0.045000%
	PSQ	ProShares	NASDAQ-100	-1x	0.9500%	0.003770%		BTC3L	MEXC	Bitcoin	+3x	36.5000%	0.100000%
	QLD	ProShares	NASDAQ-100	+2x	0.9500%	0.003770%		BTC3S	MEXC	Bitcoin	-3x	36.5000%	0.100000%
	QID	ProShares	NASDAQ-100	-2x	0.9500%	0.003770%		BTC3L	Gate.io	Bitcoin	+3x	36.5000%	0.100000%
	SQQQ	ProShares	NASDAQ-100	-3x	0.9500%	0.003770%		BTC3S	Gate.io	Bitcoin	-3x	36.5000%	0.100000%
	SPXL	Direxion	S&P500	+2x	1.0000%	0.003968%		BTC3L	AscendEX	Bitcoin	+3x	109.5000%	0.300000%
	SPXS	Direxion	S&P500	-2x	1.0800%	0.004286%		BTC3S	AscendEX	Bitcoin	-3x	109.5000%	0.300000%

5 Concluding remarks

Like leveraged ETFs, the primary goal of LVTs is to simplify investing in leveraged positions by reducing the complexities of managing such positions and limiting the risk of liquidation. During our study period, we identified numerous issues with LVTs, including a lack of transparency, custody by the issuing exchange, and possible inadequate backing. 99.9% of LVTs are implemented as centralized products, accessible only within the exchange's ecosystem. 80% of them have no interaction with the blockchain, leading to a lack of transparency in total supply, transactions, and holders.

We examined these issues along with several financial and security concerns. The total supply of 53% of LVTs is not published by the issuers, making it difficult for investors to calculate the NAV and trade LVTs at a fair market price. Additionally, 41% of LVTs may be issued with inadequate financial backing at launch, as the required futures contracts were either issued late or may not have existed at the time of the initial offering. 97% of LVTs carry the risk of front-running during well-known events, where adversaries can potentially exploit rebalancing trades. LVTs exhibit greater leverage deviation from the stated ratio compared to LETfs, due to inconsistent fund management in tokens with fixed leverage or inefficiencies in the rebalancing algorithm in LVTs with dynamic leverage.

LVTs generally have higher management fees compared to LETfs, which impacts the fund's ability to achieve its expected return. LVTs tend to underperform over extended periods (monthly or weekly) due to the compounding effect, making them unsuitable as a long-term investment vehicle.

All our findings point to the same conclusion: investors expecting simple leveraged positions that “just work” are likely to be disappointed by leveraged tokens. LVTs require careful consideration of their unique characteristics, making them more suitable for experienced traders.

Future works

Increased scrutiny from regulators and mandatory audits are potential avenues to ensure that LVTs are adequately backed. Moving LVTs on-chain could improve transparency regarding supply, transactions, and holders, while enabling self-custody. Front-running mitigation should be explored through randomized rebalancing or stealth trading (e.g., iceberg orders). LVT algorithms should be adjusted to reduce deviations from stated leverage. These measures aim to better align LVTs with investor expectations.

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A Appendices

Preview of Appendices

Our main research results are contained in the main body of the paper, but we explore a number of other issues that may also interest readers.

- Appendix A.1 describes how the return on investment (ROI) is calculated for LVTs.
- Appendix A.2 provides a comparison between LVTs and other ways of achieving leverage.
- Appendix A.3 describes how LVTs implement margin calls and forced liquidations.
- Appendix A.4 provides a measurement of LVTs offered from 2019–2023.
- Appendix A.5 describes how to invest in LVTs and its implications.
- Appendix A.6 describes deviation from states leverage in LVTs.
- Appendix A.7 describes mathematically how LVTs are re-balanced.
- Appendix A.8 compares LVT fees with LETFs.
- Appendix A.9 describes how fees in LVTs impact trading profits.
- Appendix A.10 shows how volatility reduces trading profits over time.
- Appendix A.11 illustrates a historical problematic LVT: ETCBULL offered by FTX.
- Appendix A.12 describes how LVT prices can be modeled in continuous-time.
- Appendix A.13 describes how arbitrage between an LVT and the price of underlying.

A.1 Financial leverage

Financial leverage generally refers to using derivatives (*e.g.* Options, Futures, Swaps, *etc*) or borrowed capital to boost returns. By borrowing funds, investors can amplify the size of their investment and open positions worth more than the required capital. If the investments perform well, the returns generated on the larger pool of capital can be higher compared to investing only their own funds. Borrowing also involves interest payments, which can eat into profits or exacerbate losses if investments underperform. Therefore, leveraging should be approached cautiously and with a clear understanding of the associated risks.

Financial managers of leveraged ETF (LETF) generally use derivatives mostly in the form of Total Return Swaps (TRS)³³. TRS is the predominant derivative type as they are more cost-effective and can be customized to meet specific leverage requirements. They are used to forego tax benefits related to debt or firms with relatively high business risk that avoid using large amounts of debt and paying interest on borrowing. They can also be used in a way that achieves higher returns with less upfront capital [28, 2]. The same concept is applicable to the crypto market by defining return on investment (ROI), given by:

$$ROI = R_{t_{(n-1) \rightarrow n}} = \frac{\Delta I_{t_n} - (B_{t_{n-1}}(1 + \rho_{t_{n-1}}))}{I_{t_n}} \quad (11)$$

where t_n represents the calendar time of trading hours indexed by $n = 0, 1, 2, \dots, 23$ (it can also be redefined over a different frequency such as daily, weekly, *etc*). I_{t_n} is the proceeds from the investment sale. $I_{t_{n-1}}$ is the initial investment (margin requirement). $B_{t_{n-1}}$ is the borrowed amount at the start. $\rho_{t_{n-1}}$ is the cost of borrowing at the loan time (*i.e.* accrued interest to pay).

► **Example 22.** Alice purchases one Bitcoin for \$30K at 8 am by borrowing \$20K from a crypto exchange. She deposits \$10K of her own money as a margin requirement to open the position, agreeing to a 5% interest payment on the borrowed capital. In effect, she is able to buy Bitcoin worth 3 times her own capital (3x leverage). If the price of Bitcoin increases by 10% and reaches \$33K at 10 pm, Alice's return on investment from 8 am to 10 pm, after deducting 5% interest on the borrowed amount, would be:

$$ROI = R_{t_{8 \rightarrow 22}} = \frac{(\$33K - \$10K) - (\$20K \times (1 + 5\%))}{\$10K} = \frac{\$23K - \$21K}{\$10K} = \frac{\$2K}{\$10K} = 20\%$$

With no leverage, Alice is able to purchase one-third of a Bitcoin with her \$10K. A 10% increase in the price of Bitcoin will result in a 10% profit, compared to a 20% profit when using 3x leverage (see Table 4 for the positive effects of different leverage factors).

As can be seen, Alice's profit exceeds that of Bitcoin when the market moves in her favor, but it also magnifies her losses. The amplification that leverage provides works in both directions, making it more suitable for experienced investors.

► **Example 23.** In Example 22, if the Bitcoin price falls by 10% from \$30K to \$27K, Alice's 3x leverage position loses 40% due to the combined effects of (i) the price drop, (ii) repayment

³³ Total Return Swap (TRS) is a customized swap contract between the LETF issuer and investment banks, in which they agree to exchange future positive/negative cash flows based on the returns of the underlying asset. The LETF issuer pays a fixed or floating interest rate, while the investment bank shares the total return (both gains and losses) associated with the underlying asset's performance. This allows the LETF issuer to speculate on the performance of an underlying asset without directly owning it. Interest payments provide a stable income stream for the investment banks, regardless of the underlying asset's performance.

■ **Table 4** The effect of different leverage levels on ROI for an initial investment of \$10K. The price of Bitcoin is \$30K and increases by 10% from 8 am to 10 pm.

Leverage	Buying Power	Borrowed Amount	Bitcoin Amount	Position Value (After deducting 5% Interest)	ROI	Debt Ratio
1x	\$10K	-	0.33	\$11K	10%	0%
2x	\$20K	\$10K	0.67	\$21.5K	15%	47%
3x	\$30K	\$20K	1	\$32K	20%	63%
5x	\$50K	\$40K	1.67	\$53K	30%	75%
10x	\$100K	\$90K	3.33	\$105.5K	55%	85%

of the borrowed amount, and (iii) accrued interest. Comparing position performance in Table 4 with Table 5 shows that negative returns have a greater impact than positive returns on a per unit basis.

$$ROI = R_{t8 \rightarrow 22} = \frac{(\$27K - \$10K) - (\$20K \times (1 + 5\%))}{\$10K} = \frac{\$17K - \$21K}{\$10K} = \frac{-\$4K}{\$10K} = -40\%$$

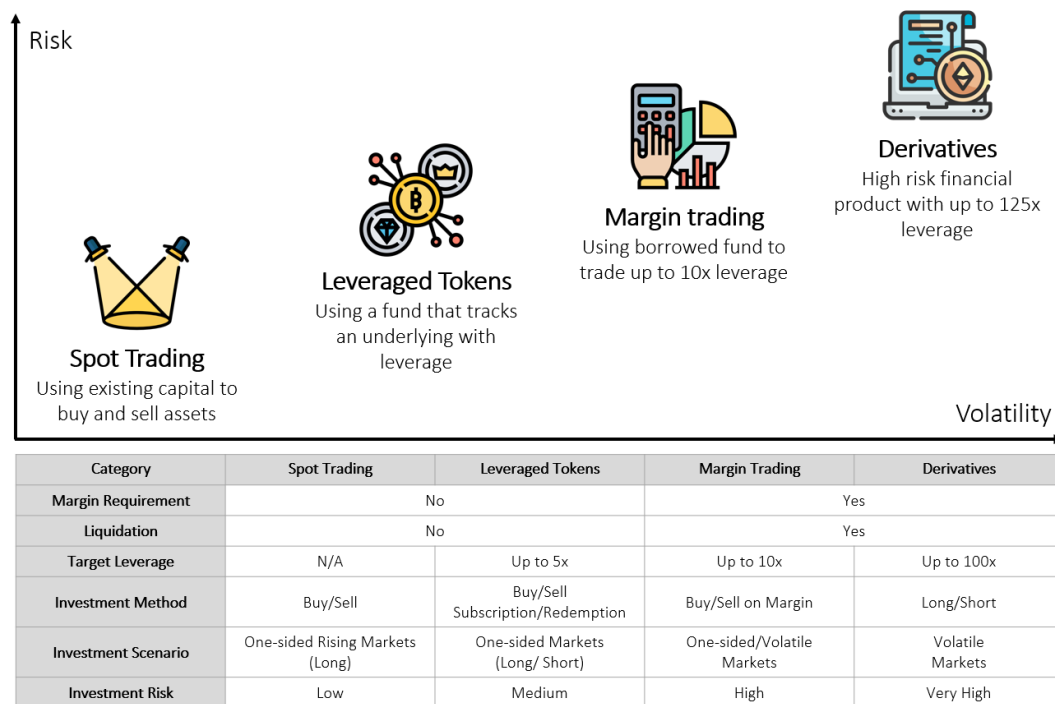
■ **Table 5** The effect of different leverage levels on the return on investment for an initial \$10K investment. The price of Bitcoin is \$30K and decreases by 10% in one day.

Leverage	Buying Power	Borrowed Amount	Bitcoin Amount	Position Value (After deducting 5% Interest)	ROI	Debt Ratio
1x	10K	0K	0.33	9K	-10%	0%
2x	20K	10K	0.67	17.5K	-25%	57%
3x	30K	20K	1	26K	-40%	77%
5x	50K	40K	1.67	43K	-70%	93%
10x	100K	90K	3.33	85.5K	-145%	105%

A.2 LVT attractiveness

There are several ways to gain leveraged exposure to crypto-assets:

- Opening a *margin account*, which allows borrowing capital and using debt, with the expectation that future gains will exceed the cost of the borrowed capital. This involves purchasing a tangible crypto asset with less capital than normally required.
- Opening a *futures position*, which may result in the acquisition of a tangible crypto asset if the contract is not perpetual and the owner opts for asset delivery upon expiration.
- Purchasing an *option contract*, which grants the owner the right to buy crypto at a strike price lower than the current spot price at expiration.
- Purchasing *leveraged tokens*, which track an underlying asset with leverage.



■ **Figure 2** Advantages of LVT compared to other trading methods

Among these options, investing in leveraged tokens (LVTs) appears more attractive. The primary advantage is the lower risk of liquidation and the simplified management of a leveraged position compared to other methods. LVTs amplify returns compared to holding only a spot position. Unlike margin trading, there is no need to borrow funds to purchase crypto-assets and subsequently pay interest. Additionally, there is no need to pledge collateral or maintain margin, as is required in margin or futures trading. The risks of LVTs are also mitigated by accounting for the effect of margin trading on potential future price crashes [50, 10]. These advantages encourage traders to purchase LVTs instead of opening positions in spot, margin, or futures markets.

However, the absence of clear standards has resulted in LVTs with varying characteristics and shortcomings. Despite the aforementioned advantages, there are risks associated with LVTs that are often overlooked by investors. In the absence of a bespoke regulatory framework for the crypto market, exchanges primarily facilitate the matching of demand and supply

(trading) [53]. LVTs are often marketed as assets with no liquidation risks, and their price is not expected to approach zero. In reality, their value gradually decreases in volatile markets, giving investors an opportunity to sell at a smaller loss.

► **Example 24.** Consider a volatile market, as shown in Table 6, where the price of Bitcoin drops 33% in one day. The BTC-Perp 3x long contract would be liquidated, while BTC3L would retain some value due to its interim rebalancing mechanism. The rebalancing process is triggered after every 10% price drop, selling part of the fund's positions. This maintains the leverage at 3x and results in a gradual price decrease compared to a sudden drop in the underlying asset. As per equation (1), the price of BTC3L at the end of the day has dropped from the initial \$10 to \$3.43, while BTC-Perp has been completely liquidated.

$$\$10 \times (1 + (3 \times -10\%)) = \$10 \times 0.7 = \$7 \Rightarrow \$7 \times 0.7 = \$4.9 \Rightarrow \$4.9 \times 0.7 = \$3.43$$

■ **Table 6** Performance of BTC-Perp and BTC3L in a volatile market where Bitcoin's price drops more than 33% in a single day. Rebalancing is triggered after each 10% drop, resulting in a gradual price decline for BTC3L, whereas BTC-Perp experiences a sharper, more immediate price drop.

	BTC-Perp					BTC3L				
	00:00	2:43	6:12	13:15	EOD	00:00	2:43	6:12	13:15	EOD
Entry Price	\$30,000	\$27,000	\$24,300	\$21,870	\$20,100	\$30,000	\$27,000	\$24,300	\$21,870	\$20,100
Price Change	0.00%	-10.00%	-10.00%	-10.00%	-3.33%	0.00%	-10.00%	-10.00%	-10.00%	-8.09%
Position Size	1	1	1	1	1	1,000	1,000	1,000	1,000	1,000
Investment Value	\$10,000	\$7,000	\$2,800	\$280	\$0	\$10,000	\$7,000	\$4,900	\$3,430	\$2,600
Profit or Lost	0%	-30%	-72%	-97%	0%	0%	-30%	-51%	-66%	-74%
Token Price						\$10.00	\$7.00	\$4.90	\$3.43	\$2.60

The above example is merely an illustration of the associated risks. One of the hidden risks is the issue of delayed or missing futures contracts. Some LVTs did not have the leveraged product available at the time of offering, while others added futures contracts to their portfolio after the initial launch. In such cases, investors were effectively purchasing a portion of a fund that was not properly backed financially.

In LVTs, leveraging via derivatives is more commonly used to increase returns on capital. However, it must be applied cautiously to avoid significant losses. Derivatives enable higher *daily* returns without requiring the full capital investment upfront³⁴. The emphasis on *daily* returns is important due to the use of short-term derivatives, which are not designed to provide the same leverage over extended periods. As a result, LVTs track their leverage objective closely over a single day but are less accurate in tracking long-term movements over weeks or months. Therefore, LVTs are particularly popular with short-term traders as they magnify the returns of potential daily market shifts [40, 6].

Another hidden risk is the issue of locking assets within crypto exchanges. Users should avoid keeping large amounts of assets on exchanges and remain informed about regulatory developments, choosing exchanges that comply with relevant regulations.

► **Example 25.** The bankruptcy of the FTX exchange in November 2022 caused LVT prices to plummet to nearly zero in three days, with users having limited opportunities to liquidate their positions through decentralized exchanges. In another instance, Binance decided in February 2024 to discontinue offering LVTs due to legal issues with the SEC and forcibly liquidated all open positions on April 3, 2024 [12].

³⁴ Returns will be slightly lower after deducting fund management fees, accounting for market volatility, interest paid on borrowing, and other associated expenses.

A.3 Liquidation risk

In most crypto exchanges, when the value of held crypto declines and the debt ratio exceeds 90%, a warning message (known as a margin call) is sent to the user, requesting additional funds to bring the account's margin level back up. If the debt ratio exceeds 95%, the exchange automatically sells the open positions to recover the borrowed capital. This process, called liquidation, is designed to protect the exchange from potential losses.

► **Example 26.** Referring to Table 5 in the previous example 23, Alice would receive a margin call if she had opened a 5x leveraged position and the price of Bitcoin dropped by 10%. Her assets are now worth \$43K, while the borrowed amount is \$40K. Only \$3K of her initial \$10K capital remains, which would not cover the borrowed amount if Bitcoin's price fell further. She would need to deposit additional funds to reduce the debt ratio below 90%, or the exchange would sell all her assets without her consent once the debt ratio reaches 95%.

Managing a margin position requires closely monitoring the underlying asset and being aware of the specific margin requirements set by the exchange to avoid potential liquidation of open positions. Alternatively, Alice could use derivatives to gain leveraged exposure to crypto assets. Unlike margin trading, users do not need to borrow when opening a futures position, although they still need to provide a fraction of the position as collateral. In futures, margin refers to the funds that parties are required to deposit into their trading accounts to enter into futures positions. It acts as insurance to ensure that both parties will meet their obligations on the delivery date and protects against price movements that may cause one party to abandon the agreement.

► **Example 27.** In the example 22, Alice could invest her \$10K initial capital in futures contracts instead of margin trading. She could benefit from higher leverage by opening, for example, a collection of 15x long Bitcoin contracts. She could buy 5 contracts with a notional value of \$30K each, totaling \$150K, with Alice only paying one-fifteenth of the notional value (\$10K). If the value of Bitcoin rises by 5%, the contracts will be worth \$31.5K each, totaling \$157.5K, giving Alice a \$7.5K profit (75% ROI). However, if the price of Bitcoin drops by 5%, the contracts will be worth \$28.5K each, and Alice's capital will shrink to \$2.5K, resulting in a 75% loss. If Bitcoin's decline continues and the price falls below \$28K, Alice's entire \$10K investment will be wiped out, and her position will be liquidated by the exchange.

This example illustrates that, unlike margin positions, where exchanges require users to increase their margin level, futures contracts do not have such a mechanism. They offer higher leverage but also come with a higher risk of liquidation. While margin positions involve borrowing money to purchase assets immediately, futures contracts are agreements to buy or sell Bitcoin at a future date. Margin positions are typically used to enhance investment returns, whereas futures are often used for speculation and hedging. Understanding these differences helps traders and investors choose the appropriate financial instruments based on their investment strategy, risk tolerance, and market outlook.

A.4 LVT statistical breakdown

The importance of studying LVTs lies in their impact on crypto investors, blockchain developers, data analysts, crypto exchanges, and external auditors. Since the exact number of issued LVTs has not yet been provided, we manually collected data from the websites of 10 LVT issuers, detailing the number of tokens issued each year. The LVTs offered by each issuer are presented separately by year and multiplier (leverage) in Figure 3. The largest number of tokens was issued in 2022, with the majority having a multiplier of three (triple-leveraged).

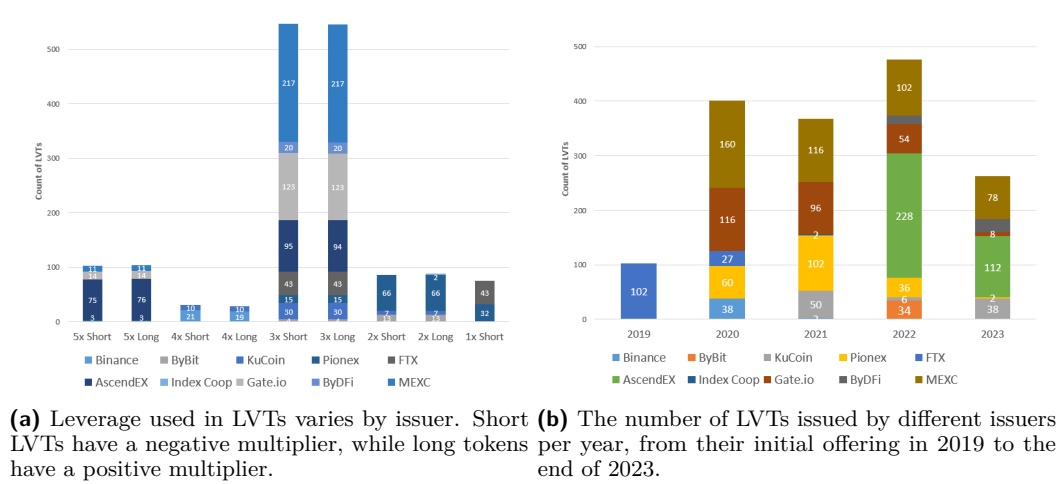


Figure 3 Comparison of LVTs issued by different issuers based on issuance date and leverage.

Crypto exchanges advertise LVTs as an investment vehicle offering leveraged exposure to crypto assets without a margin requirement and with minimal liquidation risk. However, not all LVTs are issued according to the same standard. Some are classified as cryptocurrencies with representation on the blockchain, while others are treated as derivatives available only within the crypto exchange.

Although LVT issuers attempt to compare LVTs with LETFs, the absence of a clear definition has resulted in variations in specifications and performance that often do not meet investor expectations. Despite the decentralized nature of blockchain, most LVTs are implemented centrally and are only accessible within the issuer’s ecosystem. They do not interact with decentralized applications on the blockchain, and users cannot transfer them to their crypto wallets. LVT leverage can be created through borrowing from investment banks to form a statically leveraged fund, or through derivatives (*e.g.* futures contracts) to outperform the underlying benchmark. The distinctive characteristics of LVTs, including their presence on the blockchain, the type of leverage used to amplify returns, the execution environment of the fund management algorithm, and the source of leveraged funds, are summarized in Table 2.

LVT characteristics can also be categorized in other ways. The leverage of tokens can be fixed or dynamic (variable). Depending on the issuer, the rebalancing process may be triggered at different times of the day. Some issuers perform interim rebalancing to prevent sudden drops in token prices. Furthermore, the percentage of daily expenses and the timing of deductions from the fund vary among issuers, following no uniform pattern.

A.5 LVT investment challenges

As LETFs play an increasing role in institutional portfolios over time [26], investing in LVTs could similarly expand in the cryptocurrency market. LVTs can be bought and sold (i) on the spot market like other crypto-assets, (ii) through a subscription or redemption process by dealing directly with the issuer, (iii) by interacting with a smart contract if the LVT is created on the blockchain, or (iv) using decentralized exchanges such as Kyberswap, Uniswap, Sushiswap, *etc.*

To use the spot market, users can search for token names (*e.g.* BTC3L, ETH5S, XRP2L, *etc.*) or look for *Leveraged Tokens* or *Leveraged ETF* on the top bar of the issuer's homepage. A subscription initiates an order to buy directly from the issuer, while redemption sells tokens back to the issuer. This process is generally more expensive than buying and selling on the spot market, as it affects the total supply of LVTs and requires issuer intervention. Depending on the issuer, additional charges may apply, including a 0.1% trading fee, 0.1% subscription fee, and 0.1% redemption fee. Requests are processed every 8 hours by the issuer, and if the price changes significantly, the requester may receive tokens at a different price. There is also a daily cap of \$20K to \$50K per account due to the limited supply of LVTs available each day. Therefore, using the spot market is generally a faster and more cost-effective method of buying and selling LVTs.

► **Example 28.** Alice can buy Bitcoin leveraged tokens, BTC3L or BTC3S, from the KuCoin spot market, obtaining 3x leverage without directly investing in Bitcoin. For every 1% increase in the price of Bitcoin, the value of BTC3L increases by 3% (Figure 4a), and for every 1% decrease in the price of Bitcoin, the value of BTC3S increases by 3% (Figure 4b).



(a) The daily return of the long 3x leveraged token BTC3L (orange line) issued by KuCoin is compared to the underlying Bitcoin (blue line). (b) The daily return of the short 3x leveraged token BTC3S (red line) issued by KuCoin is compared to the underlying Bitcoin (blue line).

■ **Figure 4** The daily return of both short and long leveraged tokens is compared to the daily return of the underlying asset in 2023 and 2024 (source: tradingview.com/chart).

The above example illustrates centralized LVTs, but if LVTs are created on the blockchain, it may still be possible to exchange them through the issuer's website (*e.g.* FTX tokens). The website acts as a proxy, updating balances on the smart contract in batches. Alternatively, users can send Ether directly to the smart contract to buy LVTs or exchange LVTs for stablecoins at their current value. The smart contract must expose external methods that accept cryptocurrency and update user balances after calculating the exchange rate on-chain. Smart contracts developed for LVTs typically follow the ERC-20 standard with extended

capabilities to support functions such as rebalancing, daily fee deductions, and subscription or redemption. Investing in LVTs is more complex than investing in cryptocurrencies, and the return on investment may differ from investors' expectations.

► **Example 29.** Consider the start of trading on day 1 at 12:00 AM, which lasts for 24 hours. An LVT issuer may choose the derivative market and form a *leveraged Bitcoin fund* by purchasing one long 5x BTC-Perp contract at \$6K. The fund carries a 20% risk of liquidation at \$24K by opening this 5x leverage position, paying one-fifth of Bitcoin's spot price at \$30K (cf. Appendix A.3 on liquidation dynamics). The issuer sets the initial LVT price at \$1 (P_{t_0}), resulting in 6,000 tradable BTC5L tokens at 12:00 AM. At this point, the notional value of the *leveraged Bitcoin fund* matches Bitcoin's spot price as $\$1 \times 6,000 \times 5 = \$30K$. The price of BTC5L throughout day 1 (P_{t_1}) is calculated as follows, based on the token's initial price and Bitcoin's price at 12:00 AM.

$$P_{t_1} = P_{t_0} \times \left(1 + \left(\frac{\Delta S_{t_1}}{S_{t_0}} \times 5 \right) \right) = \$1 \times \left(1 + \left(\frac{\$30,300 - \$30,000}{\$30,000} \times 5 \right) \right) = \$1.05$$

As sellers and buyers trade throughout the day, Bitcoin's price may change by 1% at 11:00 AM, causing BTC5L's price to increase by 5% to \$1.05 per token. The price of Bitcoin may then drop by 1% at 3:00 PM, eventually ending day 1 with a 2% increase at \$30,600 (S_{t_1}). This underlying price change causes BTC5L's price to first drop by 5% and then rise by 10% (P_{t_1}). S_{t_1} and P_{t_1} serve as the reference prices for calculating BTC5L's price throughout day 2. Similarly, $S_{t_{n-1}}$ and $P_{t_{n-1}}$ will be used as reference prices for calculating BTC5L's price on trading day n . For simplicity, applicable fees are omitted, and daily market fluctuations are assumed to be less than 10%, so the reference prices remain unchanged during the trading day, without triggering interim rebalancing.

This example demonstrates how LVT prices fluctuate in relation to the underlying asset and how leverage impacts returns. In general, LVT returns are path-dependent, and investors must understand short-term price movements accurately. Otherwise, recovering losses in LVTs becomes more challenging due to the compounding effect. In trending markets, this compounding can enhance gains, as positive returns build upon each other. However, in volatile or sideways markets, frequent fluctuations can lead to a decay in performance, as losses reduce the base on which future gains are calculated. Over time, this results in the leveraged token underperforming relative to the expected multiple of the underlying asset's overall return.

► **Example 30.** Consider Alice, Bob, and Charlie, each investing \$500 in the crypto market. Bob and Charlie believe they have deep market knowledge and can predict short-term price movements. Bob predicts Bitcoin will rise the next day, while Charlie predicts it will fall. Both decide to double their returns by purchasing double-leveraged BTC2L and BTC2S, respectively, instead of investing directly in Bitcoin. The return on BTC2L will be double that of Bitcoin if its price increases by 50%, and BTC2S will double if Bitcoin's price drops by 50%. Alice, being more conservative, invests directly in Bitcoin through a margin account without leverage. The next day, Bitcoin's price rises by 10%, giving Bob a 20% return compared to Alice's 10%. Charlie, on the other hand, suffers a 20% loss. After the first day, Bob holds onto his LVTs, hoping for further gains. However, on the second day, Bitcoin's price drops by 10%, causing Bob a 4% loss due to the compounding effect. The 10% price drop on the second day leads to a 20% loss in Bob's LVT, which had gained 20% the previous day. Bob incurs more significant losses because of the higher notional value of his LVTs. Charlie, meanwhile, gains no benefit from the second day's price drop, as his leveraged loss

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from the first day outweighs any potential gains. After four trading days, Bob and Charlie have lost 3.5% and 11% of their initial capital, respectively, while Alice has nearly broken even.

■ **Table 7** The compounding effect on the return of LVTs during consecutive trading days. PNLs are calculated based on the initial capital, not the previous day's notional value. Additionally, the amount borrowed by Alice illustrates the impact of interest repayment, and the \$500 could have been fully allocated by her initially.

Investor	Initial Investment	Borrowed Amount	Leverage	6% annually interest (Daily payments)	Crypto Asset		Day 1	Day 2	Day 3	Day 4
Alice	\$250.00	\$250.00	1	\$0.025	BTC		10.00%	-10.00%	-8.00%	10.00%
						Value	\$550.00	\$495.00	\$455.40	\$500.94
						PNL (\$)	\$49.98	-\$5.03	-\$44.63	\$0.92
Bob	\$500.00	\$0.00	2	\$0.000	BTC2L	PNL (%)	10.00%	-1.01%	-8.92%	0.18%
						Value	\$600.00	\$480.00	\$403.20	\$483.84
						PNL (\$)	\$100.00	-\$20.00	-\$96.80	-\$16.16
Charlie	\$500.00	\$0.00	-2	\$0.000	BTC2S	PNL (%)	20.00%	-4.00%	-19.36%	-3.23%
						Value	\$400.00	\$480.00	\$556.80	\$445.44
						PNL (\$)	-\$100.00	-\$20.00	\$56.80	-\$54.56
						PNL (%)	-20.00%	-4.00%	11.36%	-10.91%

Daily performances in Table 7 illustrate why buy-and-hold investors aiming for long-term returns should not view LVTs as suitable investment vehicles. LVTs are better suited for speculating on daily market direction. This is evident in the trading history of Bitcoin LVTs, as shown in Figure 5. The daily returns of KuCoin 3x Long Bitcoin (BTC3L) and 3x Short Bitcoin (BTC3S) are nearly mirror images in the short term, but both tokens have lost significant value over a longer period, from 2022 to 2024, due to the compounding effect. This underscores the path-dependent nature of LVT returns, which leads to the decay of gains in a buy-and-hold strategy.

Another challenge to consider is the potential for front-running in LVTs. As previously mentioned, well-known events can lead to front-running, impacting the performance of LVTs, liquidity, trading volume, and market conditions. Table 8 presents two possible scenarios for BTC5L, where someone calculates the basket delta and attempts to execute a front-running strategy. This may result in profiting from an increase in the underlying asset's price or even driving up the price of the LVT itself. These scenarios are discussed in greater detail in Section 4.4.



■ **Figure 5** Comparison of LVT returns in the short term (two images on the left) and long term (two images on the right). While long and short tokens exhibit similar returns in the short term, both lose substantial value in the long term due to the compounding effect (source: <https://tradingview.com/chart>).

■ **Table 8** The effect of front-running on the price of the underlying asset and the token during fund rebalancing is shown. See Table 12 in the appendix for more detailed calculations on the overall rebalancing process. The assumption in this example is a downtrend, where the fund initially receives funding fees from short traders and later pays funding fees as the price moves in the long direction of the token.

Initial State		No Front-running	Front-running (Scenario A)	Front-running (Scenario B)
Funding Fee (3 * 0.01%)	\$9,000	-\$9,900	-\$9,900	-\$9,900
Management Fee (0.3%)	-\$90,000	-\$99,000	-\$99,000	-\$99,000
Futures Basket	1,000	1,590.91	1,590.91	1590.91
Futures Price	\$30,000	\$33,000	\$33,330	\$33,660
Fund Value	\$30,000,000	\$52,500,030	\$53,025,030	\$53,550,030
Issued Tokens	600,000	700,000	700,000	700,000
Leverage	5.00	5.00	5.05	5.10
BTC5L Price	\$10	\$15	\$15.05	\$15.10
Tokens Value	\$6,000,000	\$10,500,000	\$10,535,000	\$10,570,000
Basket Delta	2.70	594.21		

In summary, before investing in LVTs, investors should thoroughly research and understand the product, assess the associated risks, and evaluate their investment objectives. They should also stay informed about market developments and closely monitor their positions to mitigate potential losses.

A.6 LVT leverage deviation

Another risk in LVTs is the deviation from the expected leverage ratio (also called tracking error). This can occur due to the compounding effect (as discussed in Section 4.5), volatility, fees, rebalancing costs, or other factors. This deviation can lead to discrepancies between the LVT's actual performance and the expected performance based on the stated leverage ratio. Leverage deviation is typically calculated as the standard deviation of the difference between the LVT's returns and the returns of its underlying asset, adjusted for the leverage factor. To calculate this, we first exported daily trading data from tradingview.com between May 2022 and May 2023³⁵. We then processed the data and calculated³⁶ leverage deviation as follows:

- Let P_{t_n} represents the last price at which a buyer and seller agreed to trade the asset before the market closes on trading day n .
- Let C_{t_n} is the daily change (in percentage) by comparing the closing price of the current day to the closing price of the previous trading day, given by $C_{t_n} = \frac{(P_{t_n} - P_{t_{n-1}})}{P_{t_{n-1}}}$.
- If C'_{t_n} represents the daily return of the LVT, the difference between the LVT's daily return and the target return of the underlying for each day n can be calculated as $D_{t_n} = C'_{t_n} - k \times C_{t_n}$.
- The mean of these daily return differences is given by $\bar{D} = \frac{1}{n} \sum_{t=1}^n D_{t_n}$, where n is the number of trading days.
- The variance of these differences can be calculated using $\sigma = \frac{1}{n-1} \sum_{t=1}^n (D_{t_n} - \bar{D})^2$.
- The tracking error (TE) is ultimately calculated by taking the square root of the variance, given by $TE = \sqrt{\sigma}$.

■ **Table 9** An example of the calculation of leverage deviation in BTC3L and BTC3S tokens, traded during the first two weeks of April 2023 on the ByBit exchange.

	BTC			BTC3L				BTC3S			
Date	Close	Change	Close	Change	Difference	Leverage	Close	Change	Difference	Leverage	
4/2/2023	\$ 28,155.9	-0.9819%	\$ 1.1334	-2.4193%	0.5264%	2.46	\$ 4.2336	2.9622%	0.0166%	-3.02	
4/3/2023	\$ 27,788.7	-1.3042%	\$ 1.0870	-4.0939%	-0.1814%	3.14	\$ 4.4077	4.1123%	0.1998%	-3.15	
4/4/2023	\$ 28,149.7	1.2991%	\$ 1.1290	3.8638%	-0.0334%	2.97	\$ 4.2523	-3.5256%	0.3716%	-2.71	
4/5/2023	\$ 28,154.7	0.0178%	\$ 1.1301	0.0974%	0.0441%	5.49	\$ 4.2371	-0.3575%	-0.3042%	-20.12	
4/6/2023	\$ 28,022.1	-0.4710%	\$ 1.1118	-1.6193%	-0.2064%	3.44	\$ 4.2984	1.4467%	0.0338%	-3.07	
4/7/2023	\$ 27,884.2	-0.4921%	\$ 1.0978	-1.2592%	0.2171%	2.56	\$ 4.3600	1.4331%	-0.0432%	-2.91	
4/8/2023	\$ 27,929.7	0.1632%	\$ 1.1022	0.4008%	-0.0887%	2.46	\$ 4.3430	-0.3899%	0.0996%	-2.39	
4/9/2023	\$ 28,309.7	1.3606%	\$ 1.1457	3.9467%	-0.1350%	2.9	\$ 4.1839	-3.6634%	0.4183%	-2.69	
4/10/2023	\$ 29,628.1	4.6571%	\$ 1.2950	13.0313%	-0.9398%	2.8	\$ 3.6008	-13.9368%	0.0344%	-2.99	
4/11/2023	\$ 30,179.0	1.8594%	\$ 1.3605	5.0579%	-0.5202%	2.72	\$ 3.3385	-7.2845%	-1.7063%	-3.92	
4/12/2023	\$ 29,871.0	-1.0206%	\$ 1.3214	-2.8739%	0.1878%	2.82	\$ 3.4451	3.1931%	0.1313%	-3.13	
4/13/2023	\$ 30,366.2	1.6578%	\$ 1.3789	4.3514%	-0.6219%	2.62	\$ 3.2831	-4.7023%	0.2711%	-2.84	
4/14/2023	\$ 30,438.2	0.2371%	\$ 1.3839	0.3626%	-0.3487%	1.53	\$ 3.2613	-0.6640%	0.0473%	-2.8	
					0.3859%				0.5362%		

► **Example 31.** The price fluctuations of BTC, BTC3L, and BTC3S on the ByBit exchange during the first two weeks of April 2023 are shown in Table 9. Taking the second row as an example, the price of Bitcoin futures contracts dropped by 1.3042% compared to the previous trading day. Since BTC3L and BTC3S are 3x LVTs, their values should have changed by $-1.3042\% \times 3 = -3.9126\%$ and $+1.3042\% \times 3 = +3.9126\%$, respectively. However, in reality, BTC3L's value changed by -4.0939% and BTC3S by 4.1123%. In other words, BTC3L and BTC3S deviated from their 3x stated leverage by -0.1814% and 0.1998%, respectively.

³⁵ Instructions for exporting data from tradingview.com: <https://bit.ly/4cEVWhQ>.

³⁶ Script to calculate leverage deviation: <https://bit.ly/3L1DZsL>.

While both deviations benefited investors, the fund’s value was undervalued. If investors had sold their tokens, the buyers would have acquired inflated LVTs. Notably, on April 5, there were elevated deviations, resulting in 5.49x and -20.12x returns for BTC3L and BTC3S, respectively. The standard deviation of the differences between returns over these two weeks is highlighted in the last row. It measures the variability of the difference between the LVT’s returns and its target leveraged underlying returns. A lower deviation indicates closer alignment with the target performance.

We repeated the process on the most popular LETFs and LVTs to calculate the average leverage deviations. Table 10 shows the potential impact of leverage deviation on the performance of LVTs compared to LETFs. LETFs exhibit lower and more consistent tracking errors, better aligning with the stated leverage. Conversely, LVTs show higher and more variable tracking errors, indicating less alignment with the stated leverage, which implies a higher risk of investment.

■ **Table 10** Comparison of leverage deviation in sample LVTs and LETFs from May 2022 to May 2023. A total of 268 data samples were collected from the issuers’ websites.

Market	Symbol	Issuer	Underlying Index/Asset	Direction	Target Leverage	Leverage Deviation
Equity	SPXL	Direxion	S&P500 (SPX)	Long	3	0.13%
	SPXS			Short	3	0.21%
	SSO	ProShares		Long	2	0.08%
	SDS			Short	2	0.19%
	UPRO			Long	3	0.12%
	SPXU			Short	3	0.28%
	QLD		NASDAQ-100 (NDX)	Long	2	0.07%
	QID			Short	2	0.23%
	TQQQ	Long		3	0.13%	
	SQQQ	Short		3	0.28%	
Crypto	BTC3L	MEXC	Bitcoin (BTC-Perp)	Long	3	1.22%
	BTC3S			Short	3	1.15%
	BTC3L	ByBit		Long	3	1.21%
	BTC3S			Short	3	1.10%
	BTC3L	KuCoin		Long	3	1.84%
	BTC3S			Short	3	1.58%
	BTC3L	Gate.io		Long	3	2.12%
	BTC3S			Short	3	2.10%

A.7 LVT fund rebalancing

LETFS require daily rebalancing to maintain their promoted leverage. Frequent rebalancing (re-leveraging) of LETFS during periods of high volatility is necessary to preserve leveraged exposure to the tracked index [77]. Similarly, in LVTs, each token has two distinct prices: Net Asset Value (NAV) and Market Price (or transaction price). Market Price is determined by supply and demand and does not always reflect the token's actual value. It may be higher or lower than the NAV price, which will eventually converge. Traders use NAV as a fair price to buy or sell tokens, especially in bulk. The NAV of tokens fluctuates constantly as the price of the underlying asset changes.

► **Example 32.** Consider the initial price of Bitcoin at \$30K on trading day 1. The price of Bitcoin, and consequently BTC-Perp, changes by 11.33% at 8:00 PM compared to 12:00 AM. For a 5x long leverage token BTC5L, the corresponding price change would be 56.67%. Accordingly, the NAV at 8:00 PM changes by \$5.67 from the initial \$10.00 at 12:00 AM, as given by equation (1):

$$\text{NAV}_{8pm} = \text{NAV}_{12am} \times \left(1 + \left(\frac{\$33,400 - \$30,000}{\$30,000} \times 5 \right) \right) = \$10 \times (1 + (0.1133 \times 5)) = \$15.67$$

Table 11 shows the price fluctuation of BTC5L during the first trading days at different times. The row labeled *Token Price (NAV)* demonstrates how the NAV of BTC5L tracks the price movements of BTC-Perp, changing continuously. At the end of day 1 (or the beginning of day 2), the token's value has increased by 50% compared to the 10% increase in the fund. As the price of the underlying asset changes, the total value of tokens will differ from the value represented by the fund per unit. Since the token is leveraged, its value changes proportionally to the multiplier.

■ **Table 11** Fluctuation of NAV price along with futures contracts. The rebalancing process at the beginning of day 2 adjusted the leverage by purchasing new contracts.

Time	Before Rebalancing on Day 1						Day 2
	12:00 AM	4:00 AM	8:00 AM	12:00 PM	4:00 PM	8:00 PM	12:00 AM
Futures Price	\$30,000	\$31,100	\$29,800	\$32,200	\$32,800	\$33,400	\$33,000
Futures Price Change	0.00%	3.67%	-0.67%	7.33%	9.33%	11.33%	10.00%
Token Price Change	0.00%	18.33%	-3.33%	36.67%	46.67%	56.67%	50.00%
Token Price (NAV)	\$10.00	\$11.83	\$9.67	\$13.67	\$14.67	\$15.67	\$15.00
Fund Basket Size	1,000.00	1,000.00	1,000.00	1,000.00	1,000.00	1,000.00	1,000.00
Fund Gross Value	\$30,000,000	\$31,100,000	\$29,800,000	\$32,200,000	\$32,800,000	\$33,400,000	\$33,000,000
Circulating Tokens	600,000	600,000	600,000	600,000	600,000	600,000	600,000
Tokens Value	\$6,000,000	\$7,100,000	\$5,800,000	\$8,200,000	\$8,800,000	\$9,400,000	\$9,000,000
Fund Unit Value	\$10.00	\$10.37	\$9.93	\$10.73	\$10.93	\$11.13	\$11.00
Fund Unit Value Skew	\$0.00	-\$1.47	\$0.27	-\$2.93	-\$3.73	-\$4.53	-\$4.00
Target Leverage	5.00	5.00	5.00	5.00	5.00	5.00	5.00
Real Leverage	5.00	4.38	5.14	3.93	3.73	3.55	3.67
Leverage Skew	0.00	-0.62	0.14	-1.07	-1.27	-1.45	-1.33
After Rebalancing on Day 1							
Funding Fee (3*0.01%)							-9,900.00
Management Fee (0.3%)							-99,000.00
Fee Basket Size							-3.30
Fund Basket Size (After fees)							996.70
Fund Basket Delta							366.94
Fund Basket Size (After rebalancing)							1,363.64
Actual Fund Value							\$45,000,000
Actual Leverage							5.00

► **Example 33.** In the previous example, we can define *Token value* as the *Fund unit value*, since it is a tokenized representation of the fund. The unit price of the token based on the

fund value at 8:00 PM is \$11.13, while the NAV of the token is \$15.67.

$$\text{Fund Unit Value} = \frac{\text{Fund Gross Value}}{\text{Circulating Tokens} \times \text{Target Leverage}} = \frac{\$33,400,000}{600,000 \times 5} = \$11.13$$

The alignment of these two values (*i.e.* Fund notional value versus Token notional value) is achieved through a process called *Fund rebalancing*. This process is typically scheduled daily by the *Fund management algorithm*. The value of the fund is adjusted so that its unit value matches the NAV price. If the *Leverage skew* is negative, the fund value increases by purchasing new contracts, and if the *Leverage skew* is positive, a number of fund contracts are sold to balance the value.

► **Example 34.** In the previous example, the *Leverage skew* is negative, and 366.94 new contracts will be added to the fund basket (see the *Fund Basket Delta* row in Table 11). The calculation is shown below using descriptive notation instead of formal notation, for ease of understanding:

$$\text{Tokens Value} = \text{Circulating Tokens} \times \text{Token Price (NAV)} = 600,000 \times \$15.00 = \$9,000,000$$

$$\text{Fund Gross Value} = \text{Fund Basket Size} \times \text{Futures Price} = 1,000 \times \$33,000 = \$33,000,000$$

$$\begin{aligned} \text{Fees} &= \text{Fund Gross Value} \times (\text{Management Fee} + \text{Funding Fee}) \\ &= \$33,000,000 \times (0.3\% + 0.03\%) = \$99,000 + \$9,900 = \$108,900 \end{aligned}$$

$$\text{Fund Net Value} = \text{Fund Gross Value} - \text{Fees} = \$33,000,000 - \$108,900 = \$32,891,100$$

$$\text{Target Leverage} = \frac{\text{New Fund Value}}{\text{Tokens Value}} \Rightarrow \text{New Fund Value} = \text{Tokens Value} \times 5$$

$$\begin{aligned} \text{Fund Delta} &= \text{New Fund Value} - \text{Fund Net Value} \\ &= (\$9,000,000 \times 5) - \$32,891,100 = \$12,108,900 \end{aligned}$$

$$\text{Fund Basket Delta} = \frac{\text{Fund Delta}}{\text{Futures Price}} = \frac{\$12,108,900}{\$33,000} = 366.9363$$

$$\begin{aligned} \text{Fund Basket Size} &= \frac{\text{Fund Net Value}}{\text{Futures Price}} + \text{Fund Basket Delta} \\ &= \frac{\$32,891,100}{\$33,000} + 366.9363 = 996.7 + 366.9363 = 1,363.6363 \end{aligned}$$

After rebalancing the fund positions, the *Fund basket size* will be 1,363.64 contracts, increasing the fund's value to \$45,000,000.

$$\text{Fund Value} = \text{Fund Basket Size} \times \text{Futures Price} = 1,363.6363 \times \$33,000 = \$44,999,997.9$$

$$\text{Actual Leverage} = \frac{\text{Fund Value}}{\text{Tokens Value}} = \frac{\$44,999,997.9}{\$9,000,000} = 4.9999 \simeq 5$$

Considering 600,000 tokens at \$15, the value of the tokens reaches \$9,000,000, which aligns with the 5x target leverage of the token:

$$\text{Leverage} = \frac{\text{Fund Value}}{\text{Tokens Value}} = \frac{\$45,000,000}{600,000 \times \$15} = 5$$

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■ **Table 12** The impact of the rebalancing process over 5 days on the prices of BTC and BTC5L was significant. Market conditions were volatile, with Bitcoin's price increasing by 10% on day 2, followed by declines of more than 21% on days 3 and 4. On day 5, the price recovered, rising by more than 23%.

		BTC5L - Before Rebalancing							
		Day 1	Day 2	Day 3	Day 4		Day 5		
1	Futures Price	\$30,000	\$33,000	\$32,010	\$27,208	\$25,740	\$27,594	\$31,732	\$33,003
2	Futures Price Change	0.00%	10.00%	-3.00%	-15.00%	-5.40%	7.20%	15.00%	4.01%
3	Fund Basket Size	1,000.00	1,000.00	1,363.64	1,294.52	363.06	271.20	349.82	532.29
4	Fund Gross Value	\$30,000,000	\$33,000,000	\$43,650,000	\$35,221,228	\$9,345,190	\$7,483,408	\$11,100,390	\$17,567,097
5	Funding Fee (3*0.01%)	\$9,000	-\$9,900	\$13,095	\$10,566	\$2,804	-\$2,245	-\$3,330	-\$5,270
6	Mgmt. Fee (0.3%)	-\$90,000	-\$99,000	-\$130,950	-\$105,664	-\$28,036	-\$22,450	-\$33,301	-\$52,701
7	Fund Net Value	\$29,919,000	\$32,891,100	\$43,532,145	\$35,126,130	\$9,319,958	\$7,458,712	\$11,063,758	\$17,509,125
8	Token Supply Change	0	0	50,000	-30,000	-20,000	10,000	0	-20,000
9	Circulating Tokens	600,000	600,000	650,000	620,000	600,000	610,000	610,000	590,000
10	Token Price (NAV)	\$10.00	\$15.00	\$12.75	\$3.19	\$2.33	\$3.16	\$5.54	\$6.65
11	Token Price Change	0.00%	50.00%	-15.00%	-75.01%	-26.98%	36.01%	74.98%	20.03%
12	Tokens Value	\$6,000,000	\$9,000,000	\$8,287,500	\$1,975,633	\$1,396,122	\$1,930,569	\$3,378,112	\$3,921,710
13	Target Leverage	5.00	5.00	5.00	5.00	5.00	5.00	5.00	5.00
14	Real Leverage	4.99	3.65	5.25	17.78	6.68	3.86	3.28	4.46
15	Leverage Skew	0.01	1.35	-0.25	-12.78	-1.68	1.14	1.72	0.54
		After Rebalancing							
16	Fee Basket Size	-2.70	-3.30	-3.68	-3.50	-0.98	-0.89	-1.15	-1.76
17	Fund Basket Size	997.30	996.70	1,359.95	1,291.02	362.08	270.30	348.66	530.53
18	Fund Basket Delta	2.70	366.94	-65.44	-927.96	-90.88	79.51	183.63	63.61
19	Fund Basket Size	1,000.00	1,363.64	1,294.52	363.06	271.20	349.82	532.29	594.14
20	Actual Leverage	5.00	5.00	5.00	5.00	5.00	5.00	5.00	5.00
21	Rebalancing Type	Regular	Regular	Regular	Interim	Regular	Interim	Interim	Regular
22	Rebalancing Schedule	00:00 UTC	00:00 UTC	00:00 UTC	17:43 UTC	00:00 UTC	06:12 UTC	22:32 UTC	00:00 UTC

The actual leverage might slightly differ from the target leverage (e.g., 4.9999x versus 5x) due to the deduction of fees from the fund, which impacts the net value and the number of contracts.

The above example pertains only to day 1 in a non-volatile market. To explore the rebalancing process in more depth, several different scenarios in both regular and volatile markets are presented below.

- *Day 1:* A regular day with no change in the futures price (0%).
- *Day 2:* A regular day with a 10% increase in the futures price.
- *Day 3:* A regular day with a 3% price drop and the addition of 50K tokens due to increased demand.
- *Day 4:* A volatile day with an 18% price drop, initiating interim rebalancing at 17:43.
- *Day 5:* A volatile day with a price increase of more than 23%, initiating two interim rebalancing events at 6:12 and 22:32. The token supply was also reduced by 20K.

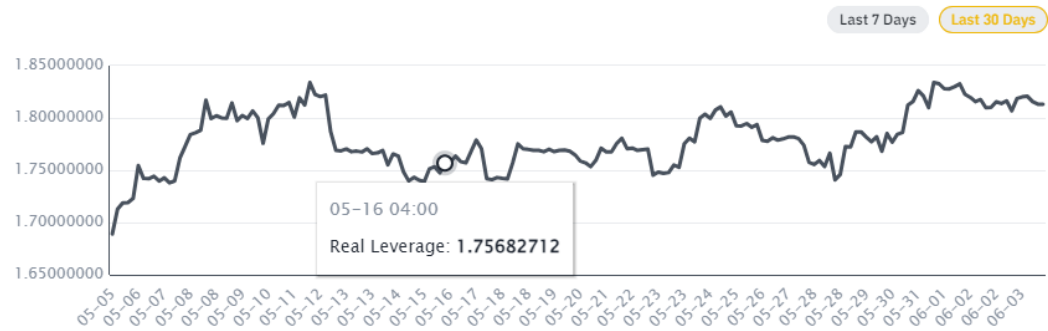
Corresponding calculations for BTC5L are provided in Table 12. To compare the impact of rebalancing on BTC5S, Table 13 shows the same daily price fluctuations. Observing the *Fund Basket Delta* row indicates that the number of contracts needed to balance the fund always aligns with the trend. In other words, if the price increases, more contracts need to be added to the fund, and if the price decreases, the number of contracts needs to be reduced. This process is reversed in BTC5S. Another important insight from Table 13 is the claim that the probability of an LVT being liquidated is nearly zero. Consider the following example:

► **Example 35.** On the fifth day of the previous example (see Day 5 column in Table 13), Alice makes two simultaneous investments. First, she buys a 5x short futures contract for \$5,518 when Bitcoin's price is \$27,594, paying one-fifth of Bitcoin's price to open the position. At the same time, she buys 674 BTC5S tokens for a total of \$5,518. On the fifth day, Bitcoin's

■ **Table 13** The impact of the rebalancing process over 5 days on the prices of BTC and BTC5S.

		BTC5S - Before Rebalancing							
		Day 1	Day 2	Day 3	Day 4		Day 5		
1	Futures Price	\$30,000	\$33,000	\$32,010	\$27,208	\$25,740	\$27,594	\$31,732	\$33,003
2	Futures Price Change	0.00%	10.00%	-3.00%	-15.00%	-5.40%	7.20%	15.00%	4.01%
3	Fund Basket Size	1,000.00	1,000.00	454.55	583.80	1,146.54	1,489.24	903.69	196.62
4	Fund Gross Value	\$30,000,000	\$33,000,000	\$14,550,000	\$15,884,083	\$29,512,017	\$41,094,049	\$28,676,045	\$6,489,012
5	Funding Fee (3*0.01%)	\$9,000	-\$9,900	\$4,365	\$4,765	\$8,854	-\$12,328	-\$8,603	-\$1,947
6	Mgmt. Fee (0.3%)	-\$90,000	-\$99,000	-\$43,650	-\$47,652	-\$88,536	-\$123,282	-\$86,028	-\$19,467
7	Fund Net Value	\$29,919,000	\$32,891,100	\$14,510,715	\$15,841,196	\$29,432,335	\$40,958,439	\$28,581,414	\$6,467,598
8	Token Supply Change	0	0	50,000	-30,000	-20,000	10,000	0	-20,000
9	Circulating Tokens	600,000	600,000	650,000	620,000	600,000	610,000	610,000	590,000
10	Token Price (NAV)	\$10.00	\$5.00	\$5.75	\$10.06	\$12.78	\$8.18	\$2.05	\$1.64
11	Token Price Change	0.00%	-50.00%	15.00%	75.01%	26.98%	-36.01%	-74.98%	-20.03%
12	Tokens Value	\$6,000,000	\$3,000,000	\$3,737,500	\$6,239,028	\$7,666,600	\$4,987,311	\$1,247,822	\$965,201
13	Target Leverage	-5.00	-5.00	-5.00	-5.00	-5.00	-5.00	-5.00	-5.00
14	Real Leverage	-4.99	-10.96	-3.88	-2.54	-3.84	-8.21	-22.91	-6.70
15	Leverage Skew	-0.01	5.96	-1.12	-2.46	-1.16	3.21	17.91	1.70
		After Rebalancing							
16	Fee Basket Size	-2.70	-3.30	-1.23	-1.58	-3.10	-4.91	-2.98	-0.65
17	Fund Basket Size	997.30	996.70	453.32	582.23	1,143.45	1,484.32	900.71	195.97
18	Fund Basket Delta	2.70	-542.15	130.48	564.32	345.79	-580.63	-704.09	-49.74
19	Fund Basket Size	1,000.00	454.55	583.80	1,146.54	1,489.24	903.69	196.62	146.23
20	Actual Leverage	-5.00	-5.00	-5.00	-5.00	-5.00	-5.00	-5.00	-5.00
21	Rebalancing Type	Regular	Regular	Regular	Interim	Regular	Interim	Interim	Regular
22	Rebalancing Schedule	00:00 UTC	00:00 UTC	00:00 UTC	17:43 UTC	00:00 UTC	06:12 UTC	22:32 UTC	00:00 UTC

price suddenly drops by more than 20%, leading to the liquidation of the 5x futures contract. Meanwhile, the interim re-leveraging of BTC5S gradually reduces its price to \$1.64 per token by the end of day 5. As a result, Alice is still able to sell her tokens for a total of \$1,105, with a cumulative loss of \$4,413 (approximately 80% capital loss). In essence, Alice lost her entire investment with the futures contracts but retained 20% of her capital by purchasing BTC5S.



■ **Figure 6** Leverage fluctuation of BTCUP (Binance long Bitcoin LVT) in May 2023.

Tables 12 and 13 show the rebalancing process of tokens with fixed leverage. The rebalancing of LVTs with variable leverage is done in the same way. Some issuers also refer to it as *fund position adjustment*. Table 14 summarizes all rebalancing details of Binance’s long Bitcoin LVT from 2021 to 2024. As can be seen, the target leverage of LVTs with dynamic leverage is random and determined at the time of rebalancing. The *Fund management algorithm* decides on a new target leverage and calculates the required futures contracts to buy or sell.

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■ **Table 14** BTCUP rebalancing history from May 2021 to April 2024, performed by Binance. The “Gap” column indicates the number of days between rebalancing events.

#	Rebalancing Time	Gap (days)	Leverage Before	Leverage After	Leverage Adjustment
1	03-Apr-2024 04:44	22	2.004	1.582	0.422
2	12-Mar-2024 02:26	5	1.740	1.529	0.211
3	06-Mar-2024 06:44	10	2.492	2.716	-0.224
4	25-Feb-2024 20:59	11	2.266	2.033	0.233
5	14-Feb-2024 11:01	24	1.872	1.980	-0.108
6	22-Jan-2024 19:28	21	2.064	2.017	0.047
7	02-Jan-2024 07:39	71	2.257	2.484	-0.227
8	23-Oct-2023 23:38	294	1.924	1.770	-0.154
9	03-Jan-2023 22:45	204	1.996	1.511	-0.485
10	13-Jun-2022 10:51	143	3.766	2.469	-1.297
11	21-Jan-2022 01:17	22	3.444	2.529	-0.915
12	30-Dec-2021 19:59	52	3.120	2.514	-0.606
13	08-Nov-2021 07:55	25	2.269	1.807	-0.462
14	14-Oct-2021 22:45	8	2.566	2.205	-0.361
15	06-Oct-2021 15:26	15	1.935	2.483	0.548
16	21-Sep-2021 20:04	11	3.033	2.519	-0.514
17	10-Sep-2021 18:37	147	3.208	2.521	-0.687
18	16-Apr-2021 12:24	0	2.029	1.956	-0.073

► **Example 36.** Consider Binance’s BTCUP token, where leverage fluctuates between 1.25x and 4x. For example, if the current leverage is 1.8x and Bitcoin’s price increases by 50%, the algorithm may set the target leverage to 2.2x (see Figure 6). Leverage is calculated as follows, targeting a random ratio within the stated range:

$$\text{Leverage} = \frac{\text{Futures Positions} \times \text{Futures Price}}{\text{Issued Tokens} \times \text{Token Price}}$$

One purpose of this random re-leveraging is to mitigate the possibility of front-running, as discussed in Section 4.4. After selecting the target leverage, the *Fund management algorithm* adjusts futures positions accordingly to align the fund’s value with the value represented by the tokens (*i.e.* matching the fund’s notional value with the tokens’ notional value).

A.8 LVT incurred fees

Similar to LETFs, there are associated expenses for LVT funds to:

- Compensate the fund management team for their expertise in constructing and managing the fund.
- Rebalance and monitor the fund, which requires active management to ensure it stays aligned with its investment objectives.
- Cover operational costs, including administrative, legal and compliance fees, custodial services, accounting, trading costs, marketing, and distribution expenses.
- Cover licensing fees if the fund replicates the performance of a specific index, requiring the licensing of the index from a third-party provider.
- Cover other applicable costs, including investor education, customer support, online account access, and related services.

The incurred fees vary depending on the fund's investment strategy, the complexity of management, services provided to investors, and liquidity. In LETFs, for example, more liquid assets tend to charge higher fees and attract short-term investors who prioritize liquidity over fees [42]. Regulatory bodies such as the SEC³⁷ regulate LETFs, focusing primarily on disclosure, transparency, investor protection, and compliance with securities laws, rather than daily management fees. Although daily fees are intended to cover associated costs, they often serve as a source of fixed income for the issuer.

► **Example 37.** On April 30, 2024, the NAV price of SPY was approximately \$501.98 per share (more precisely, \$501.980011). With SPY's expense ratio at 0.0945% per year [34], the daily expense ratio over an average of 252³⁸ trading days is $0.0945\%/252 = 0.000375\%$. The daily management fee can then be calculated as $0.000375\% \times \$501.980011 = \0.001882 per share. Given SPY's outstanding shares of 992,032,116 on that date, the total daily expenses would be $992,032,116 \times \$0.001882 = \$1,867,004$ per day (or $\$1,867,004 \times 252 = \$470,485,008$ annually).

As fees are deducted from the investment fund daily, they reduce the net return to investors. Over time, ETFs, LETFs, and LVTs with higher fees can significantly impact overall investment returns. In Table 3, the daily fees of LVTs are compared with those of LETFs to illustrate the impact of fees on investments. This information helps investors make informed decisions, balancing long-term fee stability with potential returns.

³⁷ The Securities and Exchange Commission (SEC) oversees securities exchanges, brokers, dealers, investment advisors, and mutual funds to promote fair dealing, the disclosure of important market information, and the prevention of fraud [75].

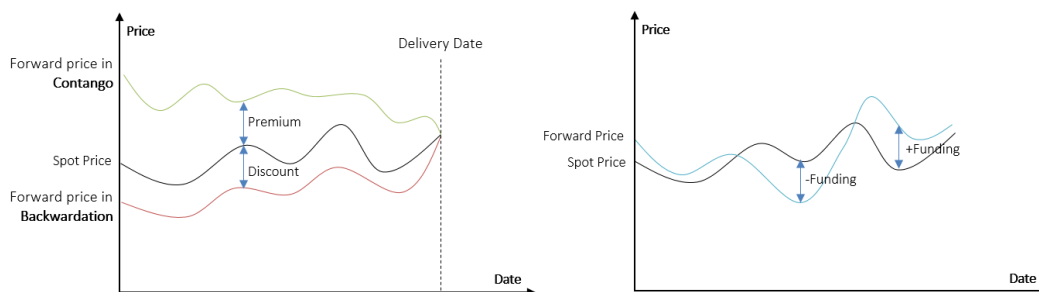
³⁸ The NYSE and NASDAQ average about 252 trading days per year. This is calculated from 365.25 (the average number of days per year) $\times 5/7$ (the proportion of workdays per week) - 6 (weekday holidays) - $4 \times 5/7$ (fixed date holidays) = 252.03 days.

A.9 LVT futures funding fee

Traditional futures contracts have an expiration date known as the delivery date. Futures prices converge with the spot price on this date. Before that, the market can be in one of two conditions: Contango or Backwardation [35]. Contango and Backwardation describe the price relationship between the spot and futures contracts. When the market is in Contango, futures are traded at a premium to the spot price (*i.e.* they are more expensive). When the market is in Backwardation, futures are traded at a discount to the spot price (*i.e.* they are cheaper than the spot price) [1]. Contango or Backwardation may occur due to shocks in supply or demand, carrying costs, geopolitical situations, pandemics, etc. Ultimately, traditional futures prices converge with the spot price by the delivery date (see the left side of Figure 7).

In crypto perpetual contracts, there is no delivery date, but the futures price still needs to settle against the spot price. At times, one side of the market becomes more aggressive, causing a disparity between spot and futures prices. To keep these prices aligned, a funding rate component is added to crypto futures. Only open positions are subject to funding payments or receipts at specific times (usually every 8 hours). If a position is closed before the funding exchange, traders do not pay or receive the funding fee. The funding fee can significantly impact high-leverage positions, potentially even leading to liquidation when paying for funding [60].

When the futures price is above the spot price, the funding rate is positive, and longs pay shorts (*i.e.* shorts receive the funding rate based on the notional value of their position, not their initial collateral). This mechanism encourages long traders to close their positions and incentivizes short traders to open new short positions. These dynamics constantly push the futures price toward the spot price and help avoid price gaps. Funding rates also reflect market sentiment and can be used to guide long or short positions. They are typically positive in a bullish market and negative in a bearish market.



■ **Figure 7** Contango or Backwardation conditions in the traditional futures market with a specific delivery date (left image). Price convergence over time in perpetual futures contracts, influenced by the funding rate (right image).

It should be noted that traders pay or receive the funding fee based on the notional value of their positions, not the price of the futures.

► **Example 38.** In Figure 8, a trader paid \$108 to open 20 lots of 5x long Bitcoin futures at \$27K³⁹. The position was opened with an average funding rate of +0.01% every 8 hours.

³⁹ In futures, a lot size is the minimum number of shares to trade. For example, the lot size of Bitcoin is 0.01 units, meaning traders can trade Bitcoin in multiples of 0.01. A 20-lot position in Bitcoin manages $20 \times 0.01 = 0.02$ Bitcoin in the futures market.

Since futures contracts always involve two sides, the trader paid \$0.054 every 8 hours to the short side, as long as the funding rate remained positive:

$$\text{Funding Fee} = \text{Notional Value} \times \text{Funding Rate} = \$108 \times 5 \times 0.01\% = \$0.054$$

The fees for large positions can be significant and may make it unprofitable to keep the position open for an extended period.

Time	Contract	Quantity(lot)	Value	Funding Rate	Funding Settlement
2023/05/30 08:00:00	BTC Perpetual/USDT	20	560.58 USDT	0.0100%	-0.05 USDT
2023/05/30 00:00:00	BTC Perpetual/USDT	20	556.58 USDT	0.0100%	-0.05 USDT
2023/05/29 16:00:00	BTC Perpetual/USDT	20	552.75 USDT	0.0100%	-0.05 USDT

■ **Figure 8** Funding fee payments every 8 hours are calculated based on the notional value of the position ($\text{Funding Fee}_{sam} = \text{Lots} \times \text{Lot Size} \times \text{Current Futures Price} \times \text{Funding Fee Rate} = 20 \times 0.01 \times \$28,029 \times 0.01\% = \$560.58 \times 0.01\% = \0.056058).

As seen in the *Realized PNL* of Figure 9, the funding fee payments every 8 hours had a negative impact on the trader's profit. The position had been open for about 5 days and had not been sold yet, but the trader had already paid a total of 0.65 USDT in funding fees.

Contract	Amount	Value	Entry Price	Mark Price	Liquidation Price	Margin (Leverage)	Unrealized PNL (ROE)	Realized PNL
BTC Perpetual/USDT	0.02 BTC	563.48 USDT	27,000.00	28,174.16	21,736.00	131.32 USDT (4.12x)	23.48 USDT (21.74%)	-0.65 USDT

■ **Figure 9** The effect of funding fee payments every 8 hours on open positions: \$23.48 unrealized PNL compared to -\$0.65 realized PNL.

The trader does not need to pay the funding fee if they close the position before the funding fee exchange schedule. The funding fee can be viewed as a cost of the contract or a penalty for keeping it open. It incentivizes traders to close positions and penalizes those who keep positions open. In simpler terms, the existence of the funding fee creates arbitrage opportunities in crypto futures, preventing futures prices from becoming inflated or deflated compared to the underlying asset price [79].

A.10 LVT volatility drag

Volatility measures the rate of change in the value of an asset. The value of a highly volatile asset can potentially spread out over a larger range of values and change dramatically over a short period of time in either direction. Conversely, the value of a low-volatility asset does not fluctuate significantly and tends to be more stable. In the equity market, for instance, when the price of a specific stock rises or falls by more than 1% over a sustained period of time, it is considered a volatile stock, especially when compared to historical price movements [38].

Volatility drag represents the rate at which the value of investments dissipates⁴⁰. As volatility increases and positions are held open for a longer period, the drag imposed by volatility accelerates, making it more destructive to the overall value of the investment [74, 58]. A similar effect occurs in LVTs, which impacts return on investment.

► **Example 39.** The performance of 3x and 5x Long/Short BTC tokens with a \$200 initial investment is compared in a volatile market.

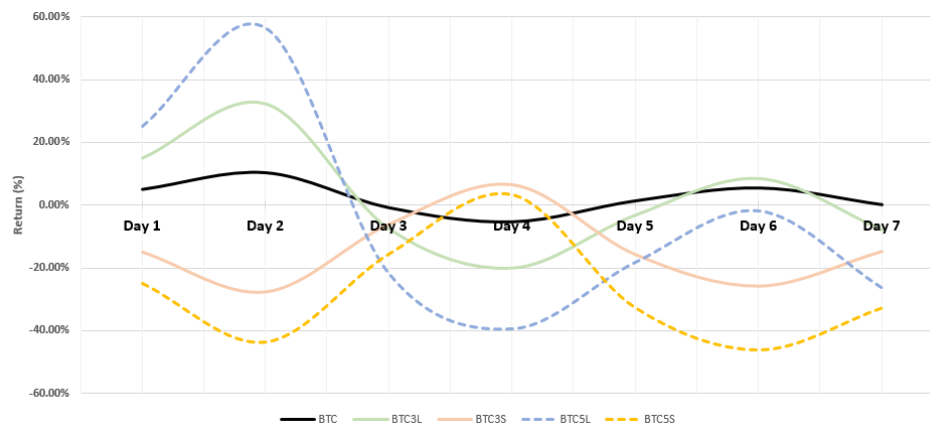
■ **Table 15** Impact of volatility drag on the performance of 3x/5x Bitcoin leveraged tokens during a volatile week. It demonstrates why LVTs are not suitable for long-term investments.

Initial Investment	Asset	Leverage		Day 1 5.0%	Day 2 5.0%	Day 3 -10.0%	Day 4 -4.5%	Day 5 7.0%	Day 6 4.0%	Day 7 -5.0%
\$200.00	BTC	1	Value	\$210.00	\$220.50	\$198.45	\$189.52	\$202.79	\$210.90	\$200.35
			PNL (\$)	\$10.00	\$20.50	-\$1.55	-\$10.48	\$2.79	\$10.90	\$0.35
			PNL (%)	5.00%	10.25%	-0.78%	-5.24%	1.39%	5.45%	0.18%
	BTC3L	3	Value	\$230.00	\$264.50	\$185.15	\$160.15	\$193.79	\$217.04	\$184.49
			PNL (\$)	\$30.00	\$64.50	-\$14.85	-\$39.85	-\$6.21	\$17.04	-\$15.51
			PNL (%)	15.00%	32.25%	-7.43%	-19.92%	-3.11%	8.52%	-7.76%
	BTC3S	-3	Value	\$170.00	\$144.50	\$187.85	\$213.21	\$168.44	\$148.22	\$170.46
			PNL (\$)	-\$30.00	-\$55.50	-\$12.15	\$13.21	-\$31.56	-\$51.78	-\$29.54
			PNL (%)	-15.00%	-27.75%	-6.07%	6.60%	-15.78%	-25.89%	-14.77%
	BTC5L	5	Value	\$250.00	\$312.50	\$156.25	\$121.09	\$163.48	\$196.17	\$147.13
			PNL (\$)	\$50.00	\$112.50	-\$43.75	-\$78.91	-\$36.52	-\$3.83	-\$52.87
			PNL (%)	25.00%	56.25%	-21.88%	-39.45%	-18.26%	-1.91%	-26.44%
	BTC5S	-5	Value	\$150.00	\$112.50	\$168.75	\$206.72	\$134.37	\$107.49	\$134.37
			PNL (\$)	-\$50.00	-\$87.50	-\$31.25	\$6.72	-\$65.63	-\$92.51	-\$65.63
			PNL (%)	-25.00%	-43.75%	-15.63%	3.36%	-32.82%	-46.25%	-32.82%

The assumption in Table 15 is a 5% BTC price increase on day 1, followed by another 5% increase on day 2, and then a 10% drop on day 3, continuing in this pattern. Mathematically, two 5% increases on days 1 and 2 should be offset by a 10% drop on day 3. However, the math does not align as expected due to accumulated profit or loss. The price of a non-leveraged BTC position increases by 5% from \$200 to \$210 on day 1 and by another 5% from \$210 to \$220.5 on day 2 (a 10.25% gain after 2 days). A 10% drop on day 3 from \$220.5 to \$198.45 results in a position that is -0.78% lower than the initial investment, making the overall return negative.

This loss is -7.43% and -21.88% of the initial investment for BTC3L and BTC5L, respectively. Investors would expect better performance from BTC3S and BTC5S on the short side. However, due to accumulated losses in the first two days, they close at -6.08% and -15.63% lower than the starting value. Even after a week, when the initial investment has nearly reached the break-even point, all LVTs remain negative regardless of their leverage and direction (See Figure 10).

⁴⁰ Also referred to as *volatility decay* or *volatility erosion*.



■ **Figure 10** LVT returns during a volatile week: the value of the initial investment (BTC in black) reached the breakeven point on day 7, but the returns of all LVTs remained negative.

As the above example suggests, volatility negatively impacts the performance of LVT investments over time, as previously discussed in the case of LETFs [32, 73]. While volatility can be beneficial in the short term, compounded daily returns produce unexpected mathematical outcomes. Investors would have made some profits if they had closed their positions on the first or second day, but holding the position through the third day would have resulted in a loss. To minimize the impact of volatility drag on LVTs, it is better to use them as short-term investments in markets with strong trends and momentum.

It is worth mentioning that the Compounding effect and Volatility drag are related but not the same. Volatility drag refers to the reduction in overall returns of a LVT due to the inherent volatility of the underlying asset. Because these tokens reset daily, fluctuations in the asset's price can erode gains over time, especially in sideways markets. Compounding effect refers to how the daily resetting of LVT impacts their long-term returns. Compounding can either magnify gains or amplify losses over time, depending on the sequence of daily returns. This effect can be positive or negative, unlike volatility drag, which is generally detrimental. In summary, while compounding can lead to either higher or lower returns due to the daily reset, volatility drag specifically highlights the adverse impact of market volatility and management fee deductions on LVT returns over time.

A.11 ETCBULL price change

ETCBULL is a 3x Long Ethereum Classic token launched by FTX in August 2019. An examination of the transaction history on the blockchain shows a sudden increase in the token supply by the FTX exchange (see Figure 11). This increase could be due to either rising demand or the issuer gradually selling tokens on the open market to raise capital.

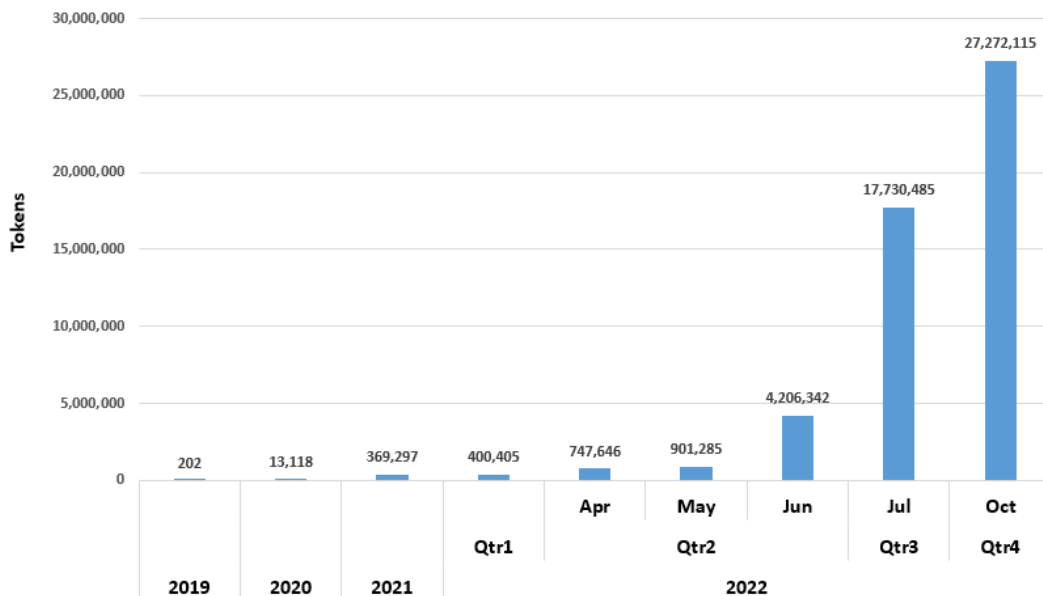


Figure 11 ETCBULL token issuance by FTX crypto exchange between Oct. 2019 and Oct. 2022 shows that 98.5% of the tokens were issued in the last 6 months of 2022, with 53% of them issued exclusively in October, just before FTX's bankruptcy in Nov. 2022.

As illustrated in Figure 12, the price of ETC remained relatively flat, fluctuating between \$22 and \$25 from May to October 2022. However, the price of ETCBULL issued by FTX dropped 4.5 times during the same period, which is not proportional to the underlying price changes of ETC.



Figure 12 ETC's price changes between May and October 2022

This bulk increase in token supply could not have been driven by new demand. In contrast, the token's price decreased nearly 4.5 times between May and October 2022 (from \$0.0255 to \$0.0057), while the underlying price remained almost flat. An analysis of historical

data shows that the average daily volume of ETCBULL was 30,000 tokens⁴¹, with most of the volume attributed to transactions between traders rather than the influx of new capital into the market.

A possible hypothesis for this devaluation could be the gradual sale of tokens by the FTX exchange to compensate for capital needs. Although this analysis is speculative, the exchange's bankruptcy in November 2022 makes this scenario more plausible. ETCBULL is not the only example; this abnormal increase in supply can also be observed in other FTX leveraged tokens. It is also possible that Binance performed a similar analysis and perceived risks associated with holding FTX tokens, which may have led to the liquidation of their FTX token holdings prior to FTX's bankruptcy [52].

⁴¹ ETCBULL historical data on Yahoo Finance: <https://yhoo.it/30zRxnf>.

A.12 LVT price modeling

The price of financial assets fluctuates due to various factors, such as market sentiment, economic news, and investor behavior. Brownian motion can be used to mathematically describe this random motion over time. Originating from physics and popularized by economists like Paul Samuelson in 1965, Brownian motion models the random movement of stock prices. It provides a way to model price fluctuations by assuming that the change in the underlying price over a short period is random and follows a normal distribution. By simulating Brownian motion, analysts and traders can generate various scenarios of how stock prices might evolve, helping them make informed decisions about investment strategies, risk management, and derivative pricing [65]. Geometric Brownian Motion (GBM) is a variant of Brownian motion, given by:

$$dS_t = \mu S_t dt + \sigma S_t dW_t \quad t \geq 0, S_0 = 1 \quad (12)$$

where S_t is the underlying price at time t , μ is the drift (expected rate of return), σ is the volatility, and dW_t is Brownian Motion (also called the Wiener⁴² process). This equation incorporates both deterministic ($\mu S_t dt$) and stochastic ($\sigma S_t dW_t$) components. The deterministic drift component μ incorporates the expected rate of return into the model, leading to a logarithmic growth pattern in the asset price. The price follows a purely random path without any deterministic trend. GBM is geometric because the drift and volatility terms are proportional to the current value of S_t . The basic properties of GBM, such as independence of increments and continuous paths, make it suitable for modeling the unpredictable nature of crypto markets [59]. To model the LVT price in continuous time, we first model the underlying price (S_t) using (12):

$$dS_t = S_t(\mu dt + \sigma dW_t) \quad (13)$$

where S_t is the underlying price at time t , dt represents a small increment of time, dS_t represents the infinitesimal change in S_t over a small time interval dt , μ is the percentage drift, where S_t drifts upwards or downwards over time, making μS_t the expected underlying return, σ is the diffusion coefficient representing the percentage of daily volatility of S_t due to unpredictable events, and dW_t is Brownian motion. This is also called a Wiener process, representing continuous random fluctuations of S_t . By incorporating equation (3) into (13), the price of LVT (P_t) can be deduced from the underlying price (S_t) as:

$$\frac{dS_t}{S_t} = \mu dt + \sigma dW_t \Rightarrow k \frac{dS_t}{S_t} = k\mu dt + k\sigma dW_t \Rightarrow dP_t = kP_t(\mu dt + \sigma dW_t) \quad (14)$$

Equation (14) suggests that LVT price (P_t) follows the same GBM as the underlying S_t , with the drift and volatility rate being k times that of S_t .

GBM is more efficient for short-term price modeling as it assumes constant volatility, while markets are prone to extreme events and possible structural breaks (*i.e.* volatility has randomness of its own). For long-term modeling, both the underlying price and volatility should be assumed to be correlated and stochastic. Therefore, stochastic volatility models such as the Heston model (developed by Steven Heston in 1993) should be used. These models assume that the volatility of the underlying asset is not constant and is presented by:

$$dS_t = S_t(\mu dt + \sqrt{v_t} dW_t) \quad (15)$$

⁴² Named in honor of American mathematician Norbert Wiener for his investigations into the mathematical properties of one-dimensional Brownian motion.

Here, the volatility ($\sqrt{v_t}$) is stochastic, meaning it evolves randomly over time. This stochastic volatility component allows the model to better capture the behavior of asset prices, especially during periods of high volatility. The Heston model describes the dynamics of both the asset price and its volatility using two stochastic differential equations: (i) modeling asset price by GBM and (ii) modeling volatility by often mean-reverting processes (*e.g.* the Cox-Ingersoll-Ross process).

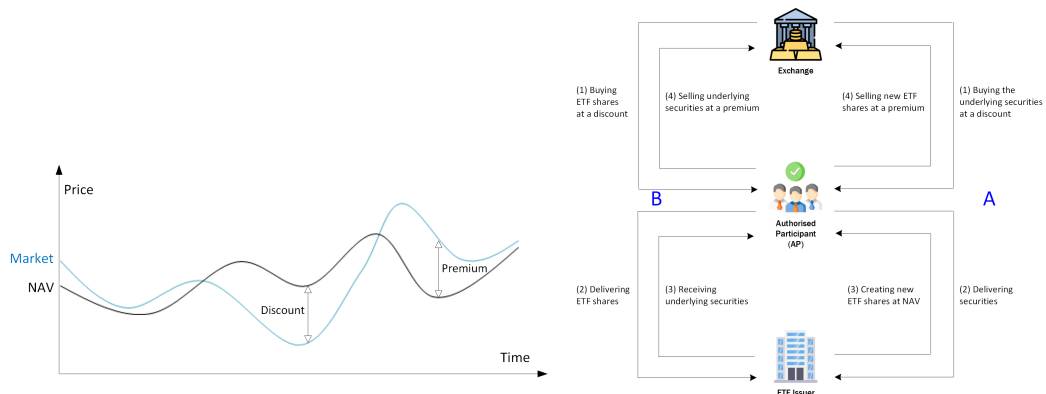
Using models that incorporate stochastic volatility (such as Heston) may provide a more accurate representation of real-world market behavior compared to GBM. However, GBM is more effective for modeling LVT prices due to their daily rebalancing process. GBM incorporates the most recent data from the previous trading day to capture new changes in market conditions. This leads to more frequent drift and volatility calibration, which aligns with the dynamic nature of LVTs. Therefore, we propose using GBM to represent LVT prices in continuous time, providing more flexibility. This model adapts quickly to new market conditions instead of generating a static forecast for a fixed period in the future.

A.13 Arbitrage mechanism

Arbitrage opportunities can arise due to non-fundamental demand shocks, which push the market price above the net asset value (NAV) [14]. The market price is the price at which shares are bought or sold on exchanges during trading hours.

► **Example 40.** On May 11, 2023, the Net Asset Value (NAV) of SPY was \$412.16, while the market price was \$412.13 per share. This indicates that SPY was trading at a discount, below NAV. Such occurrences can result from factors like market sentiment, supply and demand imbalances, dividends, expenses, or periods of market stress.

Although SPY is an ETF, it trades like shares of stocks, and its price is determined in real-time throughout the day as buyers and sellers interact. If there are more buyers than sellers, the price generally rises, and vice versa. The NAV is calculated at the end of each trading day at 4:00 p.m. Eastern Time, after the market closes. It represents the total value of assets and cash minus liabilities, divided by the number of outstanding shares. NAV serves as a benchmark to help investors understand the underlying value of assets. By comparing the NAV to the market price, investors can assess whether the asset is trading at a premium (above NAV) or a discount (below NAV) [5].



■ **Figure 13** The ETF arbitrage mechanism helps converge NAV and market prices. Authorized Participants (APs) initiate ETF creation when the ETF is trading at a premium (A) and redeem ETF shares to receive the underlying assets when the ETF is trading at a discount (B).

In ETFs, to minimize price deviations, issuers use creation and redemption mechanisms to create arbitrage opportunities, keeping market and NAV prices reasonably close (see the overall process in Figure 13). ETF issuers work with authorized participants (APs) as key players in ETF creation and redemption. Each AP (*e.g.* JP Morgan, Citigroup, etc.) has an agreement with the ETF issuer to form creation units [3].

► **Example 41.** For SPY, an AP would form a creation unit by purchasing 50,000 shares of all the S&P 500 companies. The redemption process works in reverse, where the AP would bring shares of SPY in exchange for shares of all the S&P 500 companies.

The AP can easily arbitrage discrepancies between the market value and the NAV during the trading day. If the market price rises too high compared to the NAV, the AP can step in, buy the underlying securities, and simultaneously sell shares of the ETF. Alternatively, the AP can buy shares of the ETF and sell the underlying securities if the market value falls too far below the NAV. These economic incentives provide a quick and relatively risk-free profit for the AP, while also keeping the market price and NAV closely aligned [13].